

pces application repository

(latest update August 14, 2025)

Overview

The repository github.com/iti/pcesapps provides examples of **pces** applications for reference in model building. The examples themselves are less about the applications themselves than they are about illustrating various features, attributes available to a modeler. The examples are closely tied to the discussion in [User-Extensions](#) of ways a modeler can integrate their own code into **pces** models. These examples are

- "Embedded", of an HMI computer that offloads computational tasks to an embedded system.
- "Queueing" modifies the process for initiating measurements in the same system considered by "Embedded", showing in particular how a set of simulation runs can be set up to estimate end-to-end latency as a function of the offered load of requests to the system.
- "End User Devices (EUDs)" illustrates how a user can integrate custom code to direct messages to randomly selected destinations, each of which has a different processing overhead, and measure the average end-to-end latency depending on the end user device chosen.
- "SJF" constructs a completely different system from the first three, and is constructed to show how other ways that user-written pieces of the system model can be integrated into a **pces** model. As with the EUDs application, this capability is highlighted in anticipation of users needing to keep models and code separate from public repositories such as github.
- "Flows" introduces the concept of traffic flows, as a vehicle to represent heavy traffic loads efficiently.
- "AS" shows how to include PCAP packets into the simulator, and how to add function costs for bespoke function costs on devices like routers and switches.

Before getting into the details of each model we first describe the layout of the *pcesapps* repository, and the scripts it provides to build models, and to execute them.

Repository Structure

The root directory of github.com/iti/pcesapps is called *pcesapps*, this is the directory created when the repository is copied or cloned into a user's space. Every subdirectory of *pcesapps* is dedicated to a specific application model. At the time of this writing there are three: *embedded*, *queueing*, and *sjf*, corresponding to the three examples listed above.

One intent of the overall structure is to regularize and simplify the process of creating a new model. Ideally a user need only to copy an existing application subdirectory and then replace or modify select functions, in ways that we will identify.

The structure of an application's subdirectory is likewise regularized. Each is identical, at least in the names of the files and subdirectories. These are

- *../coord/buildExp.py* , a script one calls to build the executables and the input files required to perform a set of simulation runs as an experiment. The same script can be used for each application, so we place it in a subdirectory 'coord' that is peer to the subdirectory of applications.
- *../coord/runSim.py* , a script one calls to perform the set of simulation runs defined by the model to constitute an experiment. The same script can be used for each application, so we place it in a subdirectory 'coord' that is peer to the subdirectory of applications.
- *templates*, a subdirectory that is filled by *buildExp.py* with templates of input files required for a simulation run. These templates have embedded symbols that for each simulation run are replaced with some specific value defined for it for that run.
- *input*, a subdirectory that is filled by *runSim.py* with each individual run of the experiment, with the input files that result after replacing the symbols in files found in *templates* with the values called for by that experiment.
- *output*, a subdirectory where each simulation run writes its output, and where **runSim.py* writes the summary output files, *output/results.yaml* and *output/trace.yaml*.
- *args*. This is a subdirectory that holds argument files for scripts and the simulation run itself. The contents of 'args' are
 - *args-build*, the command-line argument file for *buildExp.py*.
 - *args-run*, the command-line argument file for *runSim.py*
 - *args-sim-template* holds a set of command-line arguments that are provided to every simulation run in an experiment. In preparation for each simulation run in an experiment, *runSim.py* concatenates the contents of this file with a set of other command-line arguments, and for that run creates file *args-sim*, which is the file presented to the simulation.
- *sim-dir* is a subdirectory that holds other subdirectories for modules from which the simulation executable is built. The Docker option will build a container around this sub-directory.
- *Dockerfile-template* . The *buildExp.py* script can be directed to build a Docker container around the simulator executable. The docker process for building a container is directed by a file with hard-wired name *Dockerfile*. There may be variations in what precisely this file requires for a given application, so we have taken the approach of creating a file has the common structure, but includes symbols that are resolved and replaced for each application.
- *xlsx* is a subdirectory that primarily works as a scratch space for the call to *xlsxPCES* made from within *buildExp.py*. Starting fresh (such as what is posted to the repository), *xlsx* has a single sub-directory called *input*, and if *buildExp.py* is instructed to build a model from an *xlsxPCES* spreadsheet, that file is in *input*.

Figure 1 below graphically illustrates the structure of an application's directory. Discussion to follow will further elucidate this structure.

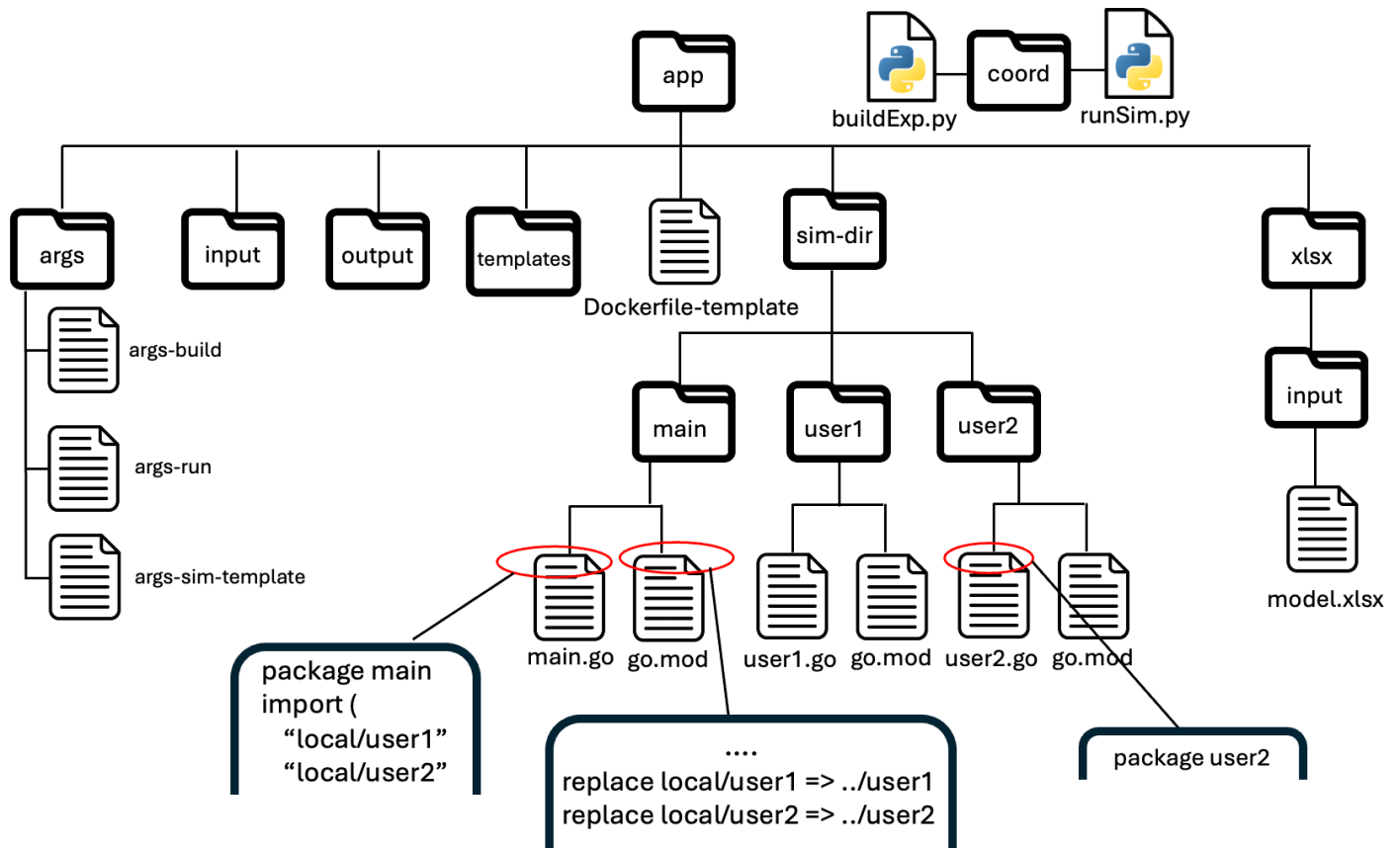


Figure 1: Structure of application directory in pcesapps repository

buildExp.py

Assuming a user has an application directory structured as in Figure 1, their first step is to decide what elements of the model are to be constructed. They edit the buildExp.py input file (args/args-build) to express their requirements, and then execute

```
% python ../coord/buildExp.py -is args/args-build
```

from the command line.

The *buildExp.py* script provides the capability of starting with an xlsxPCES spreadsheet, and going all the way to a point where an a multi-run experiment can be run, including building the simulation executable or container wrapping an executable. We imagine that a model may be built and rebuilt multiple times, particularly in response to changes the user makes to the xlsxPCES input file. The specifics of the tasks it performs are encoded in its input command-line parameter file, *args/args-build*. The fields are interpreted as follows:

Flag	Type	Use	Required
-name	string	identify project, included in output	Yes
-simexec	boolean	When included indicate that native executable should be built	No
-container	string	When included gives tag for Docker container built around simulator	No
-xpenv	string	Name of environment variable equal to path to xlsxPCES directory	No
-xlsx	string	Name of xlsxPCES input file positioned in xlsx/input subdirectory	No

Table 1: Command-line arguments for buildExp.py

If -xpenv is included, then -xlsx must also be included. Note that assignment of an environment variable typically involves adding that to a shell initiation file, and that the user must have downloaded or cloned github.com/iti/pcesbld to install the xlsxPCES tool.

Otherwise, a single call to buildExp.py may build the native executable *and* a container, or either one, or neither, and/or analyze the xlscPCES spreadsheet to create input files for a simulation experiment.

Building input files from an xlsxPCES spreadsheet

A companion document [xlsxPCES.pdf](#) describes how to use an xlsxPCES spreadsheet to describe a *pces/mrnes* model. When the -xlsx flag provides a name for an xlsxPCES input file (assumed to be in *xlsx/input*) *buildExp.py* constructs a command-line input file for xlsxPCES, and spawns a process to run the tool. The command-line input file points to subdirectory *app/templates* as the destination for its final output of templated input files for the simulator. *buildExp.py* creates scratch directories under *app/xlsx* that xlsxPCES's own command-line arguments call for. These are illustrated below.

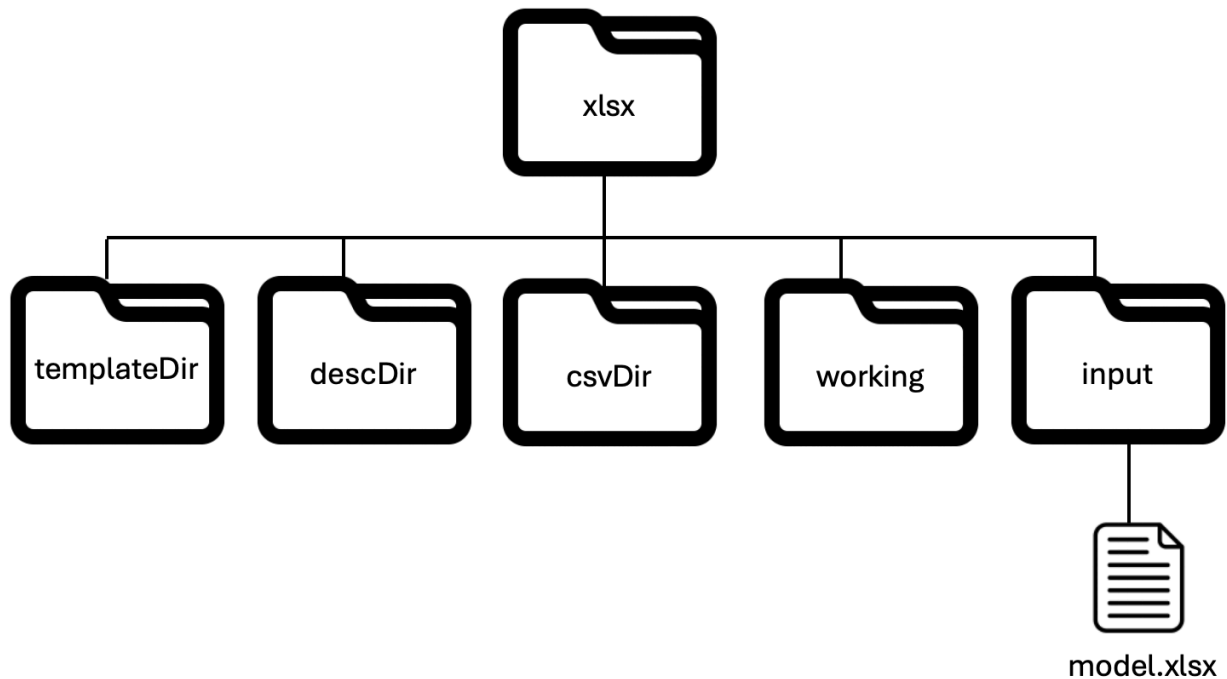


Figure 2: Structure of *xlsx* subdirectory of application directory

The roles of 'templateDir', 'descDir', 'csvDir', and 'working' are all described in [xlsxPCES.pdf](#). What is important here is that buildExp.py writes command-line arguments for xlsxPCES to position those directories under the *app/xlsx*, and positions the directory *xlsxPCES* refers to as *yamlDir*---the final output directory for templated simulation input files, as *app/templates*. The main purpose for positioning the scratch directories under *app/xlsx* is to make them easily accessible in the course of model development debugging.

Building executables

An executable is built if either *-simexec* or *-container* (or both) is present. The processing assumes a particular file structure, illustrated in Figure 1, and outlined below.

- The application's directory *sim-dir* is comprised solely of subdirectories, and (in the case of a container build), a file named *Dockerfile*.
- Each subdirectory of *sim-dir* holds the files for a go 'package' (and possibly other non-Go files), one of the subdirectories must be named *main* and contain a file called *main.go*, which is the simulation program's entry point. The first line of every Go file in a subdirectory named *subname* is "*package subname*", a statement required by Go to associate the file with the Go package whose name is the subdirectory name.
- Each subdirectory *subname* of *sim-dir* holds a file named *go.mod*, which for our file organization is required by the Go compilation process. If any *.go file in *subname* references a method or data structure in some *other* package *pckg* under *sim-dir*, then that file needs to import the package via an import statement such as `import local/pckg` and also include in *subname/go.mod* a statement `replace local/pckg => ../pckg`. This tells the Go compiler that to import package *pckg* for *subname* it should go directly to the peer subdirectory *pckg* and not try to resolve *local/pckg*. An important application of this technique is when the user has a Go package that is private, and should not be exposed to remote access.

Building a container

Docker containers are identified by tags, and the `-container` flag specifies the tag to apply to the desired container. However, it may well be that a container with the specified tag exists already. If this is the case, *buildExp.py* prints a warning that it will not create another with that same tag, advising the user to directly remove the container if a new one is required. The command for removing a container with tag TAG is

```
% docker rmi TAG
```

If no warning is returned, the command used for building the container is

```
% docker build -t TAG .
```

The `'.'` symbol references a path, to the working directory, in which a file named *Dockerfile* is to be found and interpreted by Docker.

When the intent is to make the container accessible to others, TAG may include a 'registry' url, e.g.,

Instructions in *Dockerfile* describe the steps in building the container, which define the subdirectory on the host that will appear in the container's 'space'. Critically for us, this subdirectory must include all "user code", which we mean is all the code that is compiled to create the executable in files that are not part of public repositories. User code can include Go syntax to import publically accessible modules with methods and data structures that it references, the subdirectory defining the container's space does not need to explicitly include copies or clones of those modules. This requirement drives the architecture of the file structure in Figure 1. The subdirectory *sim-dir* is the one that *Dockerfile* causes to be copied into the container.

Our main point here is that the construction steps in *Dockerfile* require knowledge of the existence and names of all the subdirectories under *sim-dir*. We don't want to require a user to have to craft by hand a correct *Dockerfile* for every application, and so *buildExp.py* automates the process, by scanning all the subdirectories in *sim-dir* and then builds a bespoke *Dockerfile* for that application, placed in *sim-dir*. The *Dockerfile-template* file in the application's main directory includes symbols that are replaced with strings derived from this analysis, with the file containing the substitutions being written into *sim-dir* as *Dockerfile*.

Building a native executable

To build the simulation executable on the host computer *buildExp.py* follows the same compilation steps as does the process for building a container. It visits every subdirectory in *sim-dir*, checks for the existence of a file *go.mod*, and calls `go mod tidy` in order to establish record of dependencies before doing the compilation itself. The main routine for the simulation executable is expected to be file *main.go* in subdirectory *main*. On passing all sanity checks the file *main.go* is compiled by

```
go build -o sim main.go
```

which creates an executable file named *sim* in the *main* subdirectory.

runSim.py

The runSim.py script coordinates the execution of multiple simulations that together constitute an 'experiment'. Its actions are defined by the settings of its command-line parameters, described below.

Flag	Use	Required
- templates	Path to directory where templated input files are stored prior to running runSim.py	Yes
-input	Path to directory where the transformed input files for a given run are written, pre-run	Yes
-output	Path to directory where simulator will write results from simulation run	Yes
-sim	Path to directory where simulation executable resides	No
-user	Boolean indicating whether a file args/args-user should also be parsed	No
-extern	Path to host directory mounted by container run command	No
- container	Tag name of container to run	No
-name	Model name, included in simulation output	Yes

Table 2: Command-line arguments for runSim.py

The meaning of these selections are given below.

- -templates, with default value being the path to the 'templates' directory illustrated in Figure 1. Each run shares many common parameters, but there is a set of parameter values that may vary from run to run. Templated input files for the experiment may contain user defined variables (or symbols), and the full set of parameters used for a simulation run is obtained by substituting concrete assignments of values defined for that run to those symbols. The result of choosing the -xpenv and -xlsx options for *buildExp.py* is to create and place those templated input files in the *templates* directory shown in Figure 1. The default value of the *runSim.py* -templates flag is the path to that directory.
- -input, with default value being the path to the 'input' directory illustrated in Figure 1. Prior to each run, runSim.py copies the files in the templates directory into the directory selected by -input, and assigns ever appearance of a symbolic experiment variable with the value defined for it, for this run. This directory serves both native execution, and container-based execution.
- -output, with default value being the path to the 'output' directory illustrated in Figure 1. Both native and container-based executions write their output to this directory.
- -sim, with default value being the 'main' directory illustrated in Figure 1. Directory where the native simulation executable is written.
- -extern, with default value (when non-empty) being the 'app' directory illustrated in Figure 1. When an experiment is run using containers, the 'docker run' command cross-mounts a directory inside of the container with the value of the '-extern' flag, a directory on the host.

- -container . When non-empty the container tag flags that the simulation runs are to be performed using a container, and the value of the flag identifies the container to run. Note that it is not necessary to have built this container with the buildExp.py step. The tag names a container registry and the name of the container within that registry. This means a service provider can have created the container and posted it to the registry, and provided the container's tag to a customer, who can write that tag as the value of the -container flag.
- -name is a label the user gives to the experiment, a string which is copied into the simulation runs' output.

Coordination of simulation runs

The input file found in *templates/experiments.yaml* describes the symbolic variables that appear in input file templates, the particular files where they are found, and for each run the value assignment to be made to that variable.

templates/experiments.yaml contains a list of dictionary entries. Each dictionary is tailored for one of the simulation runs. One of the dictionary keys is 'name', with a string value that is the unique name for the simulation run. All the other dictionary keys name a symbolic variable and a list of codes for the template input files where that symbolic variable appears. The format of one of these keys is \$symbolname, code1, code2, ..., codek , where '\$' flags that the word following is a symbolname, and each of the 'codeX' words name one of the xlsxPCES spreadsheet sheets. Each of these sheets is converted into one or two input files, and *runSim.py* maps sheet names to files as

- cp -> *cp.yaml*, *cplnit.yaml*
- execTime -> *funcExec.yaml*, *devExec.yaml*
- topo -> *topo.yaml*
- mapping -> *map.yaml*
- netParams -> *exp.yaml*
- experiments -> *experiments.yaml*

The value associated with a key is a string that for the named experiment is substituted in the indicated files for the symbol encoded in the key.

For example, imagine that *templates/experiment* contains the list


```

- $bndwdth,netParams: '100'
  $crypto,cp: AES-256-CBC
  name: exp-1
- $bndwdth,netParams: '10'
  $crypto,cp: AES-256-CBC
  name: exp-2
- $bndwdth,netParams: '100'
  $crypto,cp: AES-128-CBC
  name: exp-3
- $bndwdth,netParams: '10'
  $crypto,cp: AES-128-CBC
  name: exp-4

```

This describes an assignment of values for eight runs, named 'exp-1' through 'exp-8'. There are two symbolic variables defined, `$bndwdth`, and `$crypto`. The `$bndwdth` may appear in file *exp.yaml*, `$crypto` may appear in file *cp.yaml*. The four runs result from constructing all combinations where `\$bndwdth` is in {'10', '100'}, and `$crypto` is in {'AES-128-CBC', 'AES-256-CBC'}. The fact that all values are strings is a consequence of the way we can transform an input file template into a actualized input file by substituting one string (a value) for another (the symbolic variable).

The overall coordination of the set of runs defining an experiment is then to loop through the dictionaries in *templates/experiments.yaml*, for each preparing the run by creating the simulation app's command-line file and assigning values to symbolic variables, performing the run, and then gathering the run's output to aggregate into an overall report for the full set of runs.

Preparing for a simulation run

Every run of the simulation requires a set of command-line arguments for the simulation application. *runSim.py* prepares a command-line file for each run, as there can be variation from run-to-run (e.g., the parameter that names the run of the experiment). Many of these are set by *runSim.py* as a function of the assumed structure of Figure 1, some are drawn from a template *app/args/args-sim-template*. The command-line arguments accepted by the simulator are listed below, with some commentary.

Flag	Type	Use	Required
exptmnt	string	Name of experiment run, writting into output	Yes
inputLib	string	Path to directory with all input files	Yes
outputLib	string	Path to directory where all output is written	Yes
cp	string	Name of file with format of cp.yaml, see API	Yes
cpInit	string	Name of file with format of cpInit.yaml, see API	Yes
funcExec	string	Name of file with format of funcExec.yaml, see API	Yes
devExec	string	Name of file with format of devExec.yaml, see API	Yes
map	string	Name of file with format of map.yaml, see API	Yes
exp	string	Name of file with format of exp.yaml, see API	Yes
topo	string	Name of file with format of topo.yaml, see API	Yes
experiments	String	Name of file describing the experimental variables	Yes
trace	string	Name of file where trace is written	No
rngseed	integer	Seed for random number generator	No
stop	float	No event with event time larger than 'stop' is executed. Absence implies termination when event list is exhausted.	No
json	boolean	Flag whether input files are in .json format. Not well exercised option, probably should be deprecated.	No
csv	string	Name of file where results of measurements are written	Yes
verbose	boolean	Whether measurement output should include source and destination CmpPtn and Label information.	No
tunits	string	Units of time used to report measurements, from {nsec, musec, msec, sec}	Yes
container	boolean	Set to true when the simulation is running inside of a container	No

Table 3: Command-line arguments for simulation executable

runSim.py creates the simulation run input file and writes it into file *args/args-sim*, defining the values as follows:

- -exptmnt is taken from the input file *experiments.yaml*, where for each run the values to assign to the symbolic parameters are defined.
- -inputLib and -outputLib are the -input and -output flag values (respectively) from Table 2.
- -cp, -cplnit, -funcExec, -devExec, -map, -exp, and -topo values are 'cp.yaml', 'cplnit.yaml', 'funcExec.yaml', 'devExec.yaml', 'map.yaml', 'exp.yaml', and 'topo.yaml', respectively, being the names of files in directory *apps/input*.
- The values of -json, -trace, -rngseed, -tunits, and -stop are copied from file *args/args-sim-template*.
- -container is included if the -container flag in runSim.py's input file is non-empty, is otherwise left out (equivalent to setting to False).

runSim.py's next step is to resolve symbolic variables in input files that contain them. It can from analysis of the file *templates/experiment.yaml* determine which input files contain symbolic variables, and prior to the run creates for each a version constructed from its version in the *templates* directory one that is written into the *input* directory, using python's powerful string match-and-replace capability to substitute for each instance of the symbolic variable the concrete value specified for it this run within file *templates/experiment.yaml*.

With the construction of *args/args-sim* and resolution of symbolic variables for the input files, the run is ready to be launched.

Executing a simulation run

runSim.py uses python's 'subprocess' module to call a simulation run, either on the native host or within a container. In both cases the python code looks like

```
process = subprocess.Popen(cmdList, stdout=subprocess.PIPE, stderr=subprocess.PIPE,
text=True)
stdout, stderr = process.communicate()

if process.returncode != 0:
    print("Error from simulation run")

if len(stdout) > 0:
    print(stdout)
if len(stderr) > 0 :
    print(stderr)
```

where the call to subprocessPopen spawns a subprocess which is named with input arguments in the list 'cmdList', and establishes Unix pipes from the subprocess's stdout and stderr streams with runSim.py. A return code indicates whether the command completed successfully, and then runSim.py prints out anything the simulator pushed out through stdout or stderr.

For a run using a container 'cmdList' is assigned as

```
cmdList = ["docker", "run", "-it", "--rm", "-v", mountCmd, containerTag]
```

where `mountCmd` is a string that describes the association of an external directory with one inside the container, and `'containerTag'` identifies the container. The syntax of `'mountCmd'` is `'hostDir:containerDir'` where `hostDir` is a path to a directory on the host machine and `containerDir` is a path to a directory within the container. `runSim.py` sets `hostDir` to be the path to the `'app'` directory in Figure 1, and to `'/tmp/extern'` within the container. The presence of the `'-container'` flag in the simulation application's command-line argument file causes it to refer to `'/tmp/extern'` as the home directory for its input and output directories, and the absence of that flag causes it to refer to the directories passed to it through `'-inputDir'` and `'-outputDir'`. Otherwise, whether the simulator app is running inside or outside a container is transparent.

Sample Applications

We turn now to descriptions of several sample applications included in *pcesapps*. The examples themselves are less about the applications themselves than they are about illustrating various features, attributes available to a modeler. Three of the examples are closely tied to the discussion in [User-Extensions](#)) of ways a modeler can integrate their own code into **pces** models. These examples are

- "Embedded", of an HMI computer that offloads computational tasks to an embedded system.
- "Queuing" modifies the process for initiating measurements in the same system considered by "Embedded", showing in particular how a set of simulation runs can be set up to estimate end-to-end latency as a function of the offered load of requests to the system.
- "End User Devices" illustrates how a user can integrate custom code to direct messages to randomly selected destinations, each of which has a different processing overhead, and measure the average end-to-end latency depending on the end user device chosen. It introduces the technique of configuring the internals of **pces** to call user defining routines to respond to received messages. This capability is highlighted in anticipation of users needing to keep models and code separate from public repositories such as github.
- "Flows" gives an example where we use background flows and illustrate the effect they may have on the performance of foreground flows.

Gathering and analyzing measurements

Before diving into the details of these example applications we document how **pces** is designed to gather observations and report them. The system is built principally around the notion of measuring attributes of a path from one point in the model to another. The attribute might be latency, it might be observed available bandwidth, it might be probability of packet loss, it might be something else that the user defines. The common thread is that there is a point where the measurement of this attribute starts, and another point where it ends.

pces provides the `'measure'` class of functions to define these endpoints. Conceptually it is simple: when a message enters a `'measure'` function configured to start a measurement, the initial state and time are saved and the message is tagged. When the message is eventually delivered to a `'measurement'` function configured to end a measurement, the measurement is complete. There are details though that need attention.

Measurements between the same two points, at different times, for execution threads that all take exactly the same path between them form a natural grouping of measurements. If there is random variation among measurements in the same grouping we may wish to compute statistics on that group. We may wish to combine groupings, or further differentiate between elements in a group. **pces** supports all of these activities, meaning that a **pces** agile modeler can integrate code they developed into the **pces** framework to accomplish these tasks.

pces default behavior is to group all measurements that follow the same sequence of functions and network devices between the same two points. Its default behavior is to create a spreadsheet to summarize the measurements in a grouping, and offer to the user different options for what the spreadsheet repeats. With a single keyword in the configuration a user selects whether they want all the measurements in a group reported, the mean and variance of all measurements in the group, the mean and confidence interval around a statistical estimate of a stationary mean measurement, or a statistical range of the measurements, including minimum, maximum, 25th percent quartile, median, and 75th percent quartile.

The applications to follow demonstrate these different options.

Embedded

Application structure

Figure 3 illustrates the computational patterns (CmpPtn) for the "Embedded" example. For reference, please note that [xlsxPCES.pdf](#) goes through expression of this model in *xlsxPCES* in some detail.

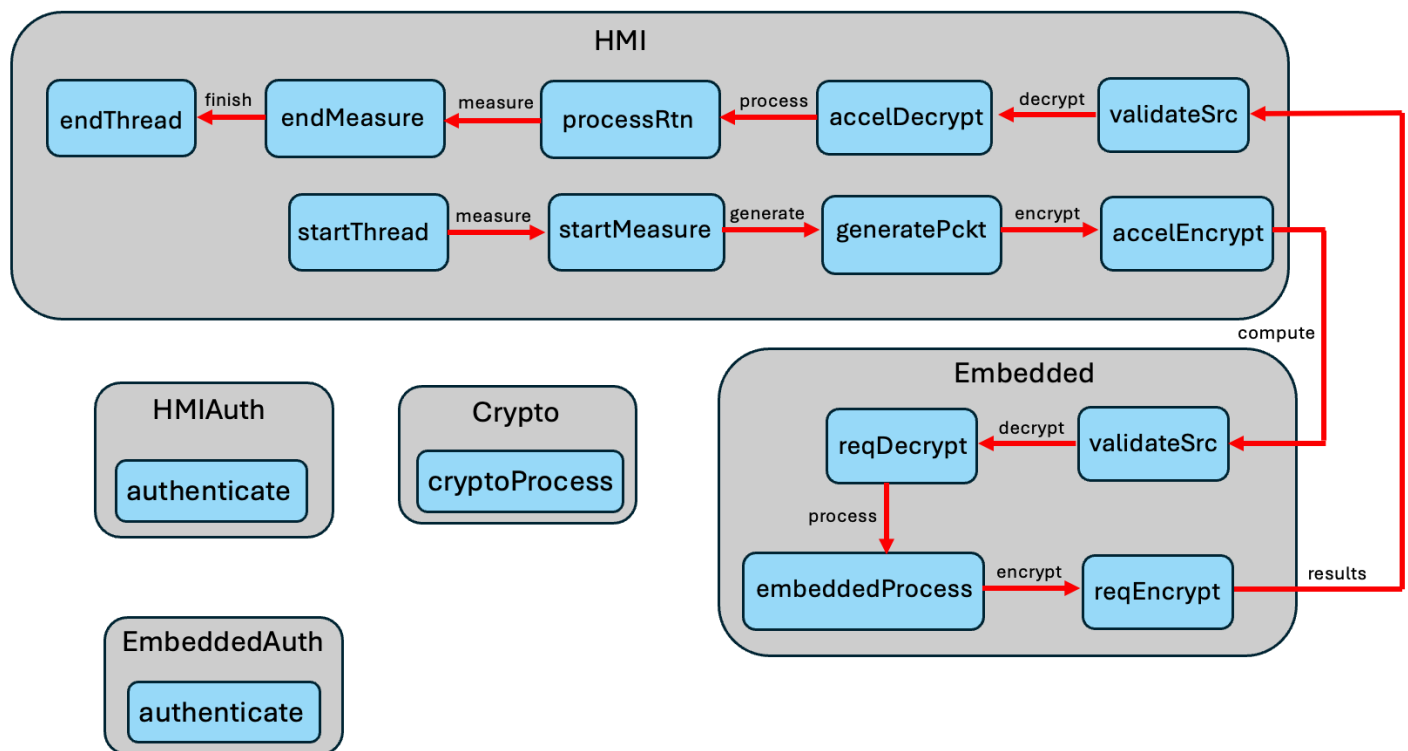


Figure 3: Computational Patterns for "Embedded" application example

Figure 4 illustrates the mapping of the CmpPtns to the topology of the example, comprised of computational Endpoints 'hmiDev', 'embeddedDev' and 'sslDev', and a switch 'hub'.

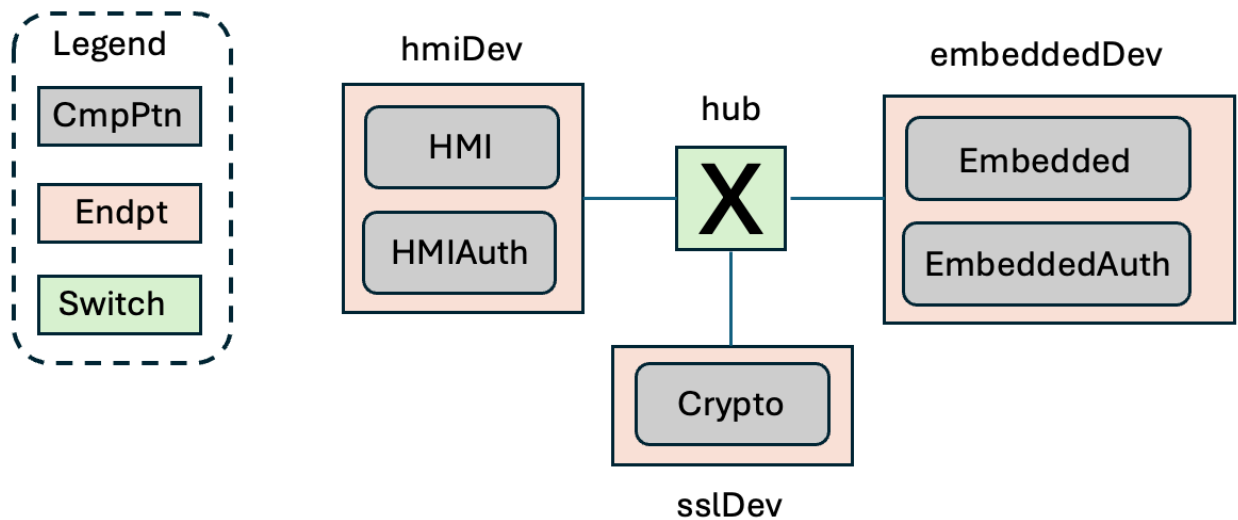


Figure 4: Topology of "Embedded example"

The big picture for us is that this workflow describes generation of a request by an HMI computer to an Embedded computer to perform some specialized computation, and have the results of that computation returned.

There are a number of details at the next layer of description. The communications between HMI and Embedded computers is encrypted, and so there are encrypt/decrypt steps to perform on both sides of a message transfer. The crypto operations on the HMI computer use an on-board hardware accelerator, while the crypto operations on the Embedded computer use an off-processor crypto server. The model represents an approach to zero-trust networking where before either computer accepts a message from the other, it first issues a challenge to the sender whose response is checked for validity before the message is accepted.

The complexities of the model are almost entirely encapsulated in the simulator's input files. These files are used principally to initialize internal simulator data structures, and then use the internal logic of the simulator to drive the model simulation. Here we show all that is needed to compile and run this model.

Application code

Figure 5 gives the code to compile, in its entirety, all of `app/sim-dir/main/main.go` in Figure 1. No additional modules are needed, subdirectories `user1` and `user2` in Figure 1 are entirely absent.

```

1 package main
2 import (
3     "github.com/iti/evt/evtm"
4     "github.com/iti/evt/vrtime"
5     "github.com/iti/pces"
6     "path/filepath"
7 )
8

```

```

 9 func main() {
10     pces.ReadSimArgs()
11     pces.RunExperiment(expCntrl, expCmplt)
12 }
13
14 func expCntrl(evtMgr *evtm.EventManager, context any, data any) any {
15     // go through all comp patterns looking for functions of the 'start' class,
16     // and schedule them for execution
17     for _, cpi := range pces.CmpPtnInstByID {
18         for _, cpfi := range cpi.Funcs {
19             if cpfi.Class == "start" {
20                 evtMgr.Schedule(cpfi, &cpfi.Class,
21                     pces.EnterFunc, vrtime.SecondsToTime(0.0))
22             }
23         }
24     }
25     return nil
26 }
27
28 func expCmplt(evtMgr *evtm.EventManager, context any, data any) any {
29     csvFileName := *context.(*string)
30     exprmntName := *data.(*string)
31
32
33     // should be only one MsrGroup
34     empty := []string{}
35     for _, msrg := range pces.MsrGrpByID {
36
37         msrg.PrepareCSVRow(csvFileName, pces.ExprmntsFile,
38             exprmntName, empty, empty, "Samples")
39
40         // write out the samples
41         msrg.AddCSVData(csvFileName, exprmntName, empty )
42         break
43     }
44
45     // put the trace file in the same directory as the msrFile
46     tdirectory, file := filepath.Split(csvFileName)
47     file = "trace.yaml"
48     traceFile := filepath.Join(tdirectory, file)
49     pces.TraceMgr.WriteToFile(traceFile, false)
50     return nil
51 }

```

Figure 5: main.go for "Embedded" application

The body of *main()* has one call to acquire and use the arguments provided in the simulator's command-line argument file. The second call (Line 11) runs the experiment, although that is split into logic that is executed before a call to routine *expCntrl*, logic executed after that call and before a call to *expCmplt()*, and logic executed after that last call.

main.go contains the two functions called from *pces.RunExperiment()*. *expCntrl()* scans every function of every *CmpPtn* looking for any that are in the 'start' class. On finding one, it schedules a call to the *pces.EnterFunc()* event handler (Lines 20-21), passing to it a pointer to the specific function, and a pointer to a string giving the function's class. The execution of the event is scheduled to occur immediately. The modeler who writes this function needs to understand what happens in this case when *pces.EnterFunc()* starts to execute. A companion document [PCES-Internals](#) describes this in detail.

expCmplt() is called after the simulation activity has completed. While it (and *expCntrl()*) share the function signature of functions that are scheduled for execution through the simulation event manager, they both are called directly. Recall from the description of *simRun.py* that the call to *expCmplt()* completes one of potentially many simulation runs that together comprise an experiment. The end result of an experiment is a spreadsheet with one more more rows describing measurements that were observed during the run. Thus we understand that what *expCmplt()* needs to do is to contribute to this spreadsheet, including (for the first run) creating the first row with column headings.

It is worthwhile pausing here to describe construction of the spreadsheet columns heading row. The heading of the first column is always "Experiment" and the cells under this heading have the experiment name for the reported data on that row. Recall the 'experiments' sheet of the *xlsxPCES* spreadsheet, these cells have the strings in its 'name' column. The heading of the second column is always 'MeasureName', and holds the common 'msrname' value (see the initialization of measure class functions in the *cp* sheet of the *xlsxPCES* spreadsheet) of the measurement endpoints associated with the group. A user may introduce additional column headers next, after which **pces** adds column headings derived from the symbol variable headings in the 'experiments' sheet of the *xlsxPCES* spreadsheet. This shows there is flexibility extended to a user in defining the output spreadsheet, and that **pces** explicitly includes cells in a row of experimental variables that may be of interest in an output analysis.

Subroutines *PrepCSVRow()* and *AddCSVRow()* are both receiver functions for a struct type *pces.MsrGroup* that holds a group's measurements. *PrepCSVRow()* prepares some **pces** data structures for the writing out of a csv row, and if the spreadsheet for the group has not yet been started, writes out the row of column headers. The last argument of the call indicates which of the output formatting options is being selected. Here "Samples" flags that all the observed measurements should be individually listed. *AddCSVRow()* appends a row to the spreadsheet. The potentially confusing 'for' loop started at Line 37 isn't really a loop, note the 'break' statement on Line 42 after one iteration. For this application the dictionary *pces.MsrGrpByID* has only one key, and the construction displayed gets at it. The key itself is a hash value; this code neither knows nor cares what the key is, only the pointer to the *pces.MsrGroup* the key maps to, so that the receiver methods can be called and produce csv rows based on the measurements it contains.

Finally, the call to **pces.TraceMgr.WriteToFile()* writes out all trace recordings saved this run, and places them into the same directory as the saved measurements.

Once all of the runs have completed, 'embedded's version of runSim.py has produced a spreadsheet written into *embedded/output/results.csv*, producing an output such as (after formatting)

Experiment	MeasureName	\$bndwidth	\$crypto	Index	Sample(msec)
exp-1	end2end	100	AES-256-CBC	1	6.01576
exp-2	end2end	10	AES-256-CBC	1	59.6558
exp-3	end2end	100	AES-128-CBC	1	6.01235
exp-4	end2end	10	AES-128-CBC	1	59.6524

The code in Figure 5 included no user defined column headers (hence the 'empty' variable passed to the receiver methods). This run was performed on the model described in *embedded/xlsx/input/embedded-model.xlsx*, which defines symbol variables *\$bndwidth* and *\$crypto*. Each experimental run named in the first column corresponds to an experimental run identified in the 'experiments' sheet of *embedded-model.xlsx*, and the values in the cells beneath them correspond to the values that sheet assigns them for each experimental run. The last columns of the output gives the measured value (default configuration is a latency), and the next-to-last column gives the position of that measurement in the group's list. This application has only one group, and only one measurement per group.

Queueing Application

Application structure

We include the "queueing" example to illustrate that a user can extend the set of command-line arguments that help govern the simulation, and in post-processing call a python program to generate a plot.

The overall structure of the queueing example is the same as the embedded example. Functionally it is different though in that the embedded example takes only one measurement per simulation run. We can imagine though experiments where the round-trip delay between two points may vary from round-trip to round-trip, as there are network objects along the execution path where queueing is introduced when needed to govern competing access to shared resources. In **mrnes** a computational endpoint where **pces** functions are executed is declared to have a particular number of cores available. If the pattern of function executions on different CmpPtns induces concurrent requests for computational service, the endpoint's bank of cores will be committed to serving as many of these as are possible, leaving others to wait in queue for service. Other points in a **pces/mrnes** model where queueing may be induced is at network interfaces where messages are pushed through the network. It is possible for multiple messages (from multiple concurrent execution threads) to converge on an interface, meaning that its bandwidth has to be shared among them or that some wait for others to receive transmission bandwidth before they do. The impact of queueing on round-trip delays can be significant under traffic patterns of sufficiently heavy load. Indeed, one of the reasons for conducting such a simulation experiment is to observe just how heavy the load must be for those impacts to be significant.

Analysis of results

As with any application posted to *pcesapps*, one builds the queueing model first, using

```
% python ../coord/buildExp.py -is args/args-build
```

and then runs the complete set of runs using

```
% python ../coord/runSim.py -is args/args-run
```

Here the *buildExp.py* script is identical to that for the embedded example, but queueing's *runSim.py* has some difference related to reporting the outcomes of the experiment.

The result of a run of the queueing model is a sequence of end-to-end latency measurements $L_1, L_2, \dots, L_k$. We aim to compute a mean latency and confidence interval around that mean. Recognizing though that the measurements are not statistically independent we adopt the 'batch-means' technique from standard simulation analysis to create samples $S_1, S_2, \dots, S_{k/b}$ that (approximately) are independent, by setting S_1 to be the average of the first b values of the original samples, S_2 be the average of the 2nd batch of b values, and so on. Since we're aiming to estimate the stationary mean latency and since the early state of the system is biased towards its initial empty state, following standard practice we discard some number d of the first batch mean samples, and use standard statistical formulas to compute the mean and confidence interval on the remaining $(k/b) - d$ batch mean samples.

There are implicitly four parameters involved here. When the measurement initiations occur close enough in time the competition among concurrent execution threads for resources will induce queuing, and so increase latency. So one of the parameters is describes the gap in time between successive generation of packets. In this model we call that parameter λ , the arrival rate, and use it as the rate parameter of an exponential random variable, treating packet generation as a Poisson process. Another parameter is the total number of measurements per run, k . A third parameter is the number of samples in a batch, b . A final parameter is the number of initial batch mean samples to drop, d . There is no obvious place in the xlsxPCES model to embed these parameters, but what we can do is support a user's inclusion of their own command-line argument file and use it to deliver these parameters to each run.

We may suppose that k , b , and d are the same for each run, and wish to observe how the latency varies as a function of λ . Recall that the xlsxPCES configuration for a start function includes a string, 'Data'. We can use that column to plug in a symbolic variable λ , changed with each run, e.g.,

### start class	cmpptn	label	pcktlen	msglen	msg type	start time	data	trace	groups
start	HMI	startThread	1000	1500	measure	0	float(λ)	FALSE	

and write into the experiments sheet something like

Experiments			
name	\$crypto,cp	\$bndwidth,netParams	\$lambda,cp
exp-1	AES-256-CBC	100	50
exp-2	AES-256-CBC	10	50
exp-3	AES-128-CBC	100	50
exp-4	AES-128-CBC	10	50
exp-5	AES-256-CBC	100	100
exp-6	AES-256-CBC	10	100
exp-7	AES-128-CBC	100	100
exp-8	AES-128-CBC	10	100

Application code

The protocol we use to enable a user-defined file of command-line arguments is to allow for the command line call to the simulator include two command-line files. When *runSim.py* script is run for the embedded example, each experiment uses the call below to run the simulator

```
% sim-dir/main/sim -is args/args-sim
```

In the case of the queueing application, *runSim.py* causes to run

```
sim-dir/main/sim -is args/args-user -is args/args-sim
```

where the additional command-line file contents are

```
-samples 10000
-batch 50
-skip 50
```

A '-user' flag in the input command file given to *runSim.py* induces the additional '-is' argument.

The '-samples' flag specifies the number of measurement initiations to make in a run, '-batch' defines the number of samples in a batch for the batch-means approach, and '-skip' specifies the number of initial batch samples to skip before doing the statistical analysis.

Of course it is user-developed code that must define what to parse in the extra file, and provision needs to be made when the command-line to the simulator is being analyzed.

Below we include *sim-dir/main/main.go* in the queueing example

```

package main
import (
    "github.com/iti/pces"
    "local/cntrl"
)

func main() {
    cntrl.ReadUserArgs(true)
    pces.ReadSimArgs()
    pces.RunExperiment(cntrl.ExpCntrl, cntrl.ExpCmplt)
}

```

Figure 6: *sim-dir/main/main.go* function for "Queueing" application

One difference from the *main.go* for embedded is the import of a module labeled as "local/cntrl", the other is a call to the *ReadUserArgs()* function in that module, critically, before the call to *pces.ReadSimArgs()*. The idea is to present the same command line to the core method for reading arguments as it would have received otherwise.

The 'cntrl' directory that is peer to 'main' (like user1 in Figure 1) holds the code below in file *uargs.go*

```

1 package cntrl
2 import (
3     "fmt"
4     "os"
5     "github.com/iti/cmdline"
6     "github.com/iti/pces"
7 )
8
9 // ReadUserArgs is called to pull off arguments from a user-defined command arguments
list
10 func ReadUserArgs(check bool) {
11     // return if either there is only one "-is" on the command-line, or
12     // if the file referenced as the user command file is not present
13     isFlags := 0
14     for idx:=1; idx< len(os.Args); idx++ {
15         if os.Args[idx] == "-is" {
16             isFlags += 1
17         }
18     }
19
20     if isFlags < 2 {
21         if check {
22             panic("expecting -is user-args on the command line")
23         } else {
24             return
25         }
26     }

```

```

27
28     userCmdFile := os.Args[2]
29     _, err := os.Stat(userCmdFile)
30     if err != nil {
31         panic(fmt.Errorf("unable to open user cmd file %s", userCmdFile))
32     }
33
34     cp := cmdline.NewCmdParser()
35     cp.AddFlag(cmdline.IntFlag, "samples", true) // nm
36     cp.AddFlag(cmdline.IntFlag, "batch", true) // nm
37     cp.AddFlag(cmdline.IntFlag, "skip", true) // nm
38
39     cp.Parse()
40
41     pces.Samples = cp.GetVar("samples").(int)
42     pces.Batch = cp.GetVar("batch").(int)
43     pces.Skip = cp.GetVar("skip").(int)
44
45     // sanity check
46     numBatches := pces.Samples/pces.Batch
47
48     if pces.Skip > numBatches/2 {
49         panic("Skipping more than 50% of the batches")
50     }
51
52     os.Args = append(os.Args[0:1], os.Args[3:]...)
53 }

```

Figure 7: *sim-dir/cntrl/uargs.go* file in "Queueing" application

lines 34-37 define what the contents of the file holding those commands is expected to look like, and lines 41-43 initialize the global variables. As it is the *pces.MsrGroup* receiver function *AddCSVRow()* that needs the variables extracted here, the variables being initialized are global in the *pces* module name space. Lines 46-50 do a sanity check to ensure that at least 1/2 of the measurements are involved in the statistical estimation. Line 52 removes "-is args/args-user" from the command-line, making *os.Args* the same as if that flag were never present.

We see in the *main.go* routine that *ExpCntrl()* and *ExpCmplt()* are for this example positioned within a new user-included module *cntrl*. The import in *main.go* of this module through *local/cntrl* is arbitrary. The go dependency file *go.mod* in *sim-dir/main* includes a line that tells the Go compiler where to find it,

```
replace local/cntrl => ../cntrl
```

i.e., in a directory named *cntrl* that is a peer to *main*.

The *ExpCntrl()* routine for queuing is different from the same function in the embedded example in that it schedules multiple executions of the measurement, as shown below.

```

1 package cntrl
2
3 import (
4
5     "github.com/iti/pces"
6     "github.com/iti/evt/evtm"
7     "github.com/iti/evt/vrtime"
8     "strconv"
9     "path/filepath"
10    "math/rand"
11 )
12
13 var seed int64 = 1324356
14
15 func interArrival(lambda float64) float64 {
16     return rand.ExpFloat64()/lambda
17 }
18
19 var saveLambda float64
20
21 // ExpCntrl is called to initialize the events for the simulation
22 func ExpCntrl(evtMgr *evtm.EventManager, context any, data any) any {
23     rand.Seed(seed)
24
25     // get the lambda parameter
26     cpi := pces.CmpPtnInstByName["HMI"]
27     cpfi := cpi.Funcs["startThread"]
28
29     srtcfg := cpfi.Cfg.(*pces.StartCfg)
30     lambda, err := strconv.ParseFloat(srtcfg.Data, 64)
31     if err != nil {
32         panic(err)
33     }
34
35     if !(lambda > 0.0) {
36         panic("expect lambda to be positive float")
37     }
38
39     saveLambda = lambda
40
41     arrivalTime := interArrival(lambda)
42     for idx:=0; idx<pces.Samples; idx++ {
43         evtMgr.Schedule(cpfi, &cpfi.Class, pces.EnterFunc,
vrtime.SecondsToTime(arrivalTime))
44         arrivalTime += interArrival(lambda)

```

```

45     }
46     return nil
47 }

```

Figure 8: ExpCntrl function in sim-dir/cntrl/exp.go of "Queueing" application

We will be computing randomly distributed intervals of time between successive initiation of packet generation, and initialize the Go system's random number generator with a known seed in line 23. This is for demonstration purposes, a more sophisticated way of including random number seeds would be expected in a useable application. Lines 26 and 27 acquire a pointer to the (known) function in the (known) CmpPtn where the initiations occur, knowledge of the names of the CmpPtn and function is leveraged here. In line 29 The *Cfg* field of the function struct is of type *any*, but we know the actual type (*pces.StartCfg*) from looking at file *pces/classes.go*, and we know that the *Data* field of that struct is a string where we placed the *\$lambda* parameter. Line 39 saves the value for use in crafting an output string later, and then lines 42 through 45 schedule 'Samples' number of initiations, with the simulation time between successive ones sampled from an exponential distribution with rate 'lambda'. In this application all of the execution threads whose launch is scheduled here will travel between the same two measurement functions, taking the same path, and so the resulting measurements will all be gathered together in one group.

The user command arguments we showed earlier initialized *pces.Samples* to 10,000 initiations. The steps in *ExpCntrl* preload the event list with every initiation that will occur. Since the cost of doing an operation in an event list increases with the size of the list, this pre-loading of events is not as efficient as a technique we will see in the "EUDs" application to follow.

cntl.ExpCmplt() is called when the simulation run ends. Comparison with the code of Figure 5 shows that it is exactly the same as *expCmplt* for the 'Embedded' application, save that here the last argument of *PrepCSVRow* is *CI*---signifying a batch-means statistical analysis with estimated mean and confidence interval, rather than a listing of all samples.

```

48 // ExpCmplt is called at the end of a simulation run, to aggregate statistics
49 // on the measurements
50 func ExpCmplt(evtMgr *evtm.EventManager, context any, data any) any {
51     csvFileName := *context.(*string)
52     exprmntName := *data.(*string)
53
54     // create a csv header
55     empty := []string{}
56
57     // get the lone group and build the csv row
58     for _, msrg := range pces.MsrGrpByID {
59         msrg.PrepareCSVRow(csvFileName, pces.ExprmntsFile,
60             exprmntName, empty, empty, "CI")
61         msrg.AddCSVData(csvFileName, exprmntName, empty)
62         break
63     }
64

```

```

65    // put the trace file in the same directory as the msrFile
66    tdirectory, file := filepath.Split(csvFileName)
67    file = "trace.yaml"
68    traceFile := filepath.Join(tdirectory, file)
69    pces.TraceMgr.WriteToFile(traceFile, false)
70    return nil
71 }

```

Figure 9: Listing of ExpCmplt() function for 'Queuing' application

Formatted, the resulting spreadsheet for the model stored in *queueing/xlsx/input/queueing.xlsx* is shown below. It gives a confidence interval around the mean time (in milliseconds) required for a round-trip of a probe packet.

Experiment	MeasureName	\$bndwidth	\$crypto	\$lambda	Samples	Mean(msec)	CI
exp-1	end2end	100	AES-256-CBC	50	350	6.00761	0.00396981
exp-2	end2end	10	AES-256-CBC	50	350	59.9973	0.095638
exp-3	end2end	100	AES-128-CBC	50	350	6.00761	0.00396981
exp-4	end2end	10	AES-128-CBC	50	350	59.9973	0.095638
exp-5	end2end	100	AES-256-CBC	100	350	6.00734	0.00466231
exp-6	end2end	10	AES-256-CBC	100	350	61.1511	0.19496
exp-7	end2end	100	AES-128-CBC	100	350	6.00734	0.00466231
exp-8	end2end	10	AES-128-CBC	100	350	61.1511	0.19496

What we learn from this is that the non-crypto costs of the model completely dominate the impact of changing the crypto block size, there is essentially no difference between the run-time costs when the block size is 256 than when it is 128, all other factors being the same. We see that there is small but measurable difference in average round-trip time between the case when we launch 50 measurements per second and 100 measurements per second. This difference can be attributed to the queueing. There is a huge difference though in whether the interface bandwidth speeds are 10Mbps or 100Mbps.

EUDS Application

Application structure

One of **pces** early users expressed interest in a reference architecture such as that illustrated in Figure 10. The beige rectangles denote computational endpoints, 'pvtHub' and 'pubHub' are switches, 'pvtRtr' and 'pubRtr' are routers, and the grey boxes carry names of CmpPtns that have functions on the computational endpoints.

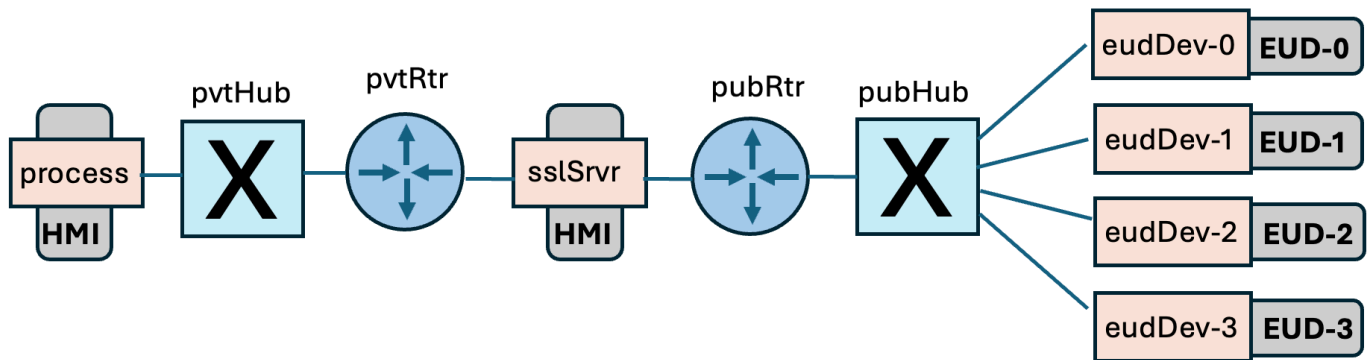


Figure 10: Architecture for study of system performance of network of End User Devices

The workflow has packets generated at a 'source' processor, be encrypted by special purpose hardware on an 'sslSrvr' device, be carried to some End User Device (EUD) where processing is done, a result is computed and is returned back to the source processor. As we've learned by now, a **pces** model required many details. We describe this workflow with $N+1$ instances of CmpPtns, where N is the number of EUDs in the model. The pattern of functions and messages is the same for each of the EUD CmpPtn's, but they do encapsulate functions that are executing on different processors. The CmpPtns and their functions are illustrated in Figure 11.

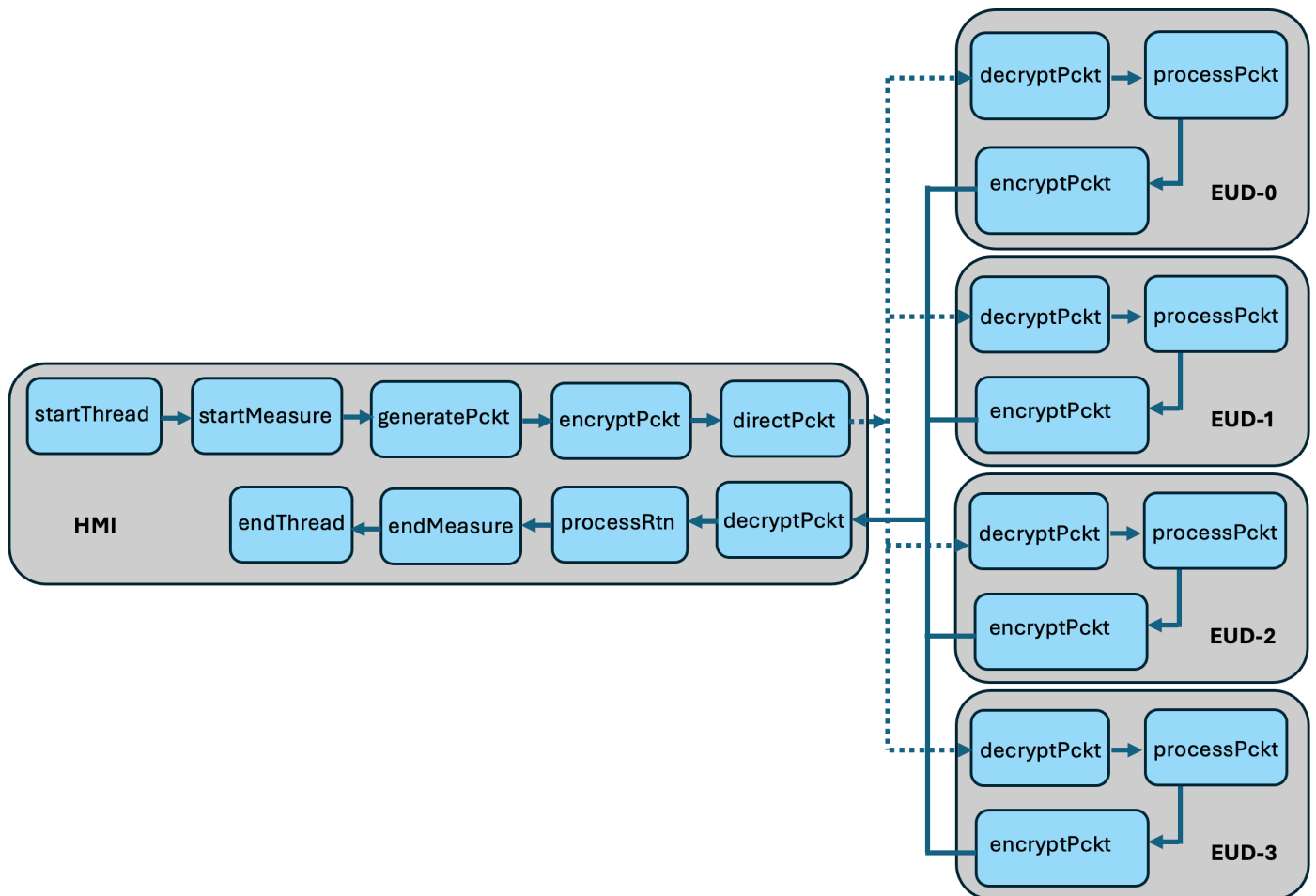


Figure 11: CmpPtns and functions for the EUDs example

The workflow of the HMI CmpPtn resembles the CmpPtn of the same name in the 'Embedded' example. When a measurement is initiated the 'startThread' function starts a new execution thread, the 'startMeasure' function begins a measurement, the 'generatePckt' function creates a packet and pushes it along to the 'encryptPckt' function for encryption. The next function in line, 'directPckt', is new to our discussion. 'The next' step for a packet is to be delivered to a function on an EUD that decrypts it, does some processing, encrypts a result, and sends it back. However, we have multiple EUDs; in this example there are four, there could be many many more. One approach would be to have the the last function in the HMI CmpPtn define an external edge to every 'decryptPckt' function in every EUD. The destination EUD will have to be selected somewhere earlier in the workflow, and a search be done on 'directPckts' list of out edges. When the number of EUDs is large that search could be time-consuming (from the point of view of the simulator overhead).

'directPckt' is a function of the 'transfer' class. It can be configured to extract the next CmpPtn, function label, and message type code from the message it is passed, which is the approach we've taken with this example. We will show how to initiate a thread with a custom routine associated with the 'startThread' that does what the default entry routine does, but also selects an EUD destination and embeds that information in the message it creates. The message carries that through all the function simulations up to 'directPckt', which uses it to direct the message to one of the EUD CmpPtns. There, as we see in Figure 6, the message is decrypted and some processing cost is applied. That cost can be different for each EUD, and in the saved example it is. The simulator accounts then for the cost of encrypting result the, and then directs (through an external edge defined in the configuration) the message back to the HMI CmpPtn. The effects of decryption, processing the return, completing the measurement, and finishing the execution thread are then all accomplished.

Application code

The code changes between this example and "Queueing" are noticeable at initialization. Figure 12 lists the new one's version of the startup routine *ExpCntrl()*. Compare with Figure 8.

```

27 func ExpCntrl(evtMgr *evtm.EventManager, context any, data any) any {
28     rand.Seed(seed)
29
30     // get the lambda parameter
31     cpi := pces.CmpPtnInstByName["HMI"]
32     cpfi := cpi.Funcs["startThread"]
33
34     // check validity
35     srtcfg := cpfi.Cfg.(*pces.StartCfg)
36     lambda, err := strconv.ParseFloat(srtcfg.Data, 64)
37     if err != nil || !(lambda > 0.0) {
38         panic("expect lambda to be positive float")
39     }
40
41     // kick off the first initiation, after one inter-initiation time
42     arrivalTime := interArrival(lambda)
43     SaveLambda = lambda
44     // schedule the first start

```

```

45     evtMgr.Schedule(cpfi, nil, startEnterCycle,
46         vrtime.SecondsToTime(arrivalTime))
47
48     // initialize data structures focus on EUD CmpPtns
49     eudCPID = make([]int,4)
50     eudCPID[0] = pces.CmpPtnInstByName["EUD-0"].ID
51     eudCPID[1] = pces.CmpPtnInstByName["EUD-1"].ID
52     eudCPID[2] = pces.CmpPtnInstByName["EUD-2"].ID
53     eudCPID[3] = pces.CmpPtnInstByName["EUD-2"].ID
54
55     // order the list for repeatability
56     sort.Ints(eudCPID)
57
58     // compute demonstration probabilities of EUD target
59     eudPr = make(map[int]float64)
60     eudPr[eudCPID[0]] = 0.1
61     eudPr[eudCPID[1]] = 0.3
62     eudPr[eudCPID[2]] = 0.6
63     eudPr[eudCPID[3]] = 1.0
64
65     // put in a custom path classifier
66     cpfi = cpi.Funcs["endMeasure"]
67     state := cpfi.State.(*pces.MeasureState)
68     state.Classify = eudClassify
69
70     // make a map of cpfi.ID to cpfi.PtnName
71     fID2CP = make(map[int]string)
72
73     cpi    = pces.CmpPtnInstByName["EUD-0"]
74     cpfi   = cpi.Funcs["eudProcess"]
75     fID2CP[cpfi.ID] = "EUD-0"
76
77     cpi    = pces.CmpPtnInstByName["EUD-1"]
78     cpfi   = cpi.Funcs["eudProcess"]
79     fID2CP[cpfi.ID] = "EUD-1"
80
81     cpi    = pces.CmpPtnInstByName["EUD-2"]
82     cpfi   = cpi.Funcs["eudProcess"]
83     fID2CP[cpfi.ID] = "EUD-2"
84
85     cpi    = pces.CmpPtnInstByName["EUD-3"]
86     cpfi   = cpi.Funcs["eudProcess"]
87     fID2CP[cpfi.ID] = "EUD-3"
88
89     return nil
90 }

```

Figure 12: Body of ExpCntrl routine in sim-dir/cntrl/exp.go of "EUDS" application

Lines 27-40 get the arrival rate parameter of a Poisson stream of measurement initiations, much as before. Lines 45-46 schedule a new routine, *startEnterCycle()* after an inter-arrival delay, and unlike the 'Queueing' example, only one of them. We'll return to this routine after exploring the rest of the *ExpCntrl()*. Lines 44-88 initialize data structures that aid in the selection of a target EUD, and in reporting at the time the measurement is completed which of the EUDs was visited on its route. We'll soon see that we select a EUD at random, from a probability distribution that is not uniform. Lines 49-63 construct a map whose index is the ID of the EUDs and whose value is the probability that that EUD or one that appears earlier than it in the sorted list of EUD ids is selected. Routine *startEnterCycle()* will use this data structure.

Lines 70-87 creates a map *fID2CP* we use to classify a measurement. When a message enters the measurement function that completes the measurement, the full sequence of IDs of functions and network devices that were visited on its path is available. We want to determine which of the EUDs is on that sequence, and on finding it, have a string we can use for output that identifies it. So the *fID2CP* dictionary maps the function ID of one of the EUD functions---*eudProcess*---and assigns as dictionary value the name of the *CmpPtn* that function is mapped to.

We turn our attention now to *startEnterCycle()*. Recalling the logic of function initiation, the execution of *pces.EnterFunc()* looks up a response function for the function pointed to (here, *startThread*), the default being *pces/class.go:enterStart()*. The code below in Figure 13 was created by copying that routine to be included in *sim-dir/cntrl/exp.go*, renamed from *startEnter* to *startEnterCycle*, and edited to include random sampling of an EUD destination.

```

88 func startEnterCycle(evtMgr *evtm.EventManager, context any, data any) any {
89     cpfi := context.(*pces.CmpPtnFuncInst)
90     srts := cpfi.State.(*pces.StartState)
91     cpm := new(pces.CmpPtnMsg)
92
93     pces.NumExecThreads += 1
94     cpm.ExecID = pces.NumExecThreads
95
96     cpm.PcktLen = srts.PcktLen
97     cpm.MsgLen = srts.MsgLen
98
99     // randomly select which next EUD to generate an arrival for
100    u := rand.Float64()
101    cpid := 0
102    for idx:=0; idx<len(eudCPID); idx++ {
103        if u < eudPr[eudCPID[idx]] {
104            cpid = idx
105            break
106        }
107    }
108
109    // get the CPID of the next EUD to target
110    cpm.XCPID = eudCPID[cpid]
111
112

```

```

113     srts.Calls    += 1
114     cpm.XLabel    = "decryptPckt"
115     cpm.XMsgType  = "decrypt"
116
117     endptName := cpfi.Host
118     endpt := mrnes.EndptDevByName[endptName]
119     pces.AddCPTrace(pces.TraceMgr, cpfi.Trace, evtMgr.CurrentTime(),
120         cpm.ExecID, endpt.DevID(),
121         pces.FullFuncName(cpfi,"startEnterCycle"), cpm)
122
123     // out edge destination a function of the message type
124     cpm = pces.AdvanceMsg(cpfi, cpm, srts.MsgType)
125
126     cpfi.AddResponse(cpm.ExecID, []*pces.CmpPtnMsg{cpm})
127     evtMgr.Schedule(cpfi, cpm, pces.ExitFunc, vrtime.SecondsToTime(0.0))
128
129     // schedule the next invocation of startEnterCycle after arrivalTime amount of time
130     if srts.Calls < Samples {
131         arrivalTime := interArrival(SaveLambda)
132         evtMgr.Schedule(cpfi, nil, startEnterCycle, vrtime.SecondsToTime(arrivalTime))
133     }
134     return nil
135 }

```

Figure 13: Body of *startEnterCycle* routine in *sim-dir/cntrl/exp.go* of "EUDS" application

Lines 89-98 are essentially copies of lines in *startEnter*. Lines 99-107 use a standard technique for sampling discrete random variables by sampling a uniform 0-1 random variable *u* and searching a cumulative probability distribution list for the first element that dominates it. Line 110 records the ID of the selected EUD in the XPCPDI field of the message, and lines 114-115 set the (known) function label and message type to have 'directPckt' place on its outbound message. Lines 117-126 are copied from *startEnter*. The scheduling at line 127 is slightly different in that *pces.ExitFunc* is scheduled to executed immediately, whereas the structure of *startEnter* includes an initial delay that we don't use in this example. Lines 130-133 are new to *startEnterCycle* and serve to schedule the next call to *startEnterCycle* based on a sampled inter-arrival delay, but to do so if the number of initialiations has not yet passed the user-included number of samples carried in global variable 'Samples'.

As we have described, the group to which a measurement belongs depends on the path the message takes. While in this particular application we know at the point of initiation how many different paths there will be, and indeed use that knowledge to build data structures to aid in reporting which path was taken, we are demonstrating here a general approach of recording the trajectory and analyzing it to determine the group.

Figure 14 provides the code for the 'EUDS' application's version of the *ExpCmplt()* routine.

```

154 // ExpCmplt is called at the end of a simulation run, to aggregate statistics
155 // on the measurements
156 func ExpCmplt(evtMgr *evtm.EventManager, context any, data any) any {

```

```

157     csvFileName := *context.(*string)
158     exprmntName := *data.(*string)
159
160     // create a csv header
161     hdr := []string{"EUD"}
162     empty := []string{}
163
164     // cycle through groups
165     for _, msrg := range pces.MsrGrpByID {
166         msrg.PrepareCSVRow(csvFileName, pces.ExprmntsFile,
167             exprmntName, hdr, empty, "Range")
168
169         data := []string{msrg.GroupDesc}
170         msrg.AddCSVData(csvFileName, exprmntName, data)
171     }
172
173     // put the trace file in the same directory as the msrFile
174     tdirectory, _ := filepath.Split(csvFileName)
175     file := "trace.yaml"
176     traceFile := filepath.Join(tdirectory, file)
177     pces.TraceMgr.WriteToFile(traceFile, false)
178     return nil
179 }

```

Figure 14: Body of *ExpCmplt()* routine in *sim-dir/cntrl/exp.go* of "EUDS" application

This looks very much like the ones we've seen for the other applications, with two exceptions. First, the last argument of *PrepareCSVRow()* is selecting that the 'Range' representation of the data be given as output. Secondly, Line 161 creates a user-defined column heading, intended to report which of the EUDs was visited by the measurements in a group, and that header is included by line 167 into the spreadsheet columns. Line 169 defines the value of the row's cell in that column. Here the 'GroupDesc' holds a string determined by analyzing a measurement's path with the user defined function *eudClassify*.

```

92 var fID2CP map[int]string
93 func eudClassify(visited []int) string {
94     for _, devID := range visited {
95         name, present := fID2CP[devID]
96         if present {
97             return name
98         }
99     }
100     return "Default"
101 }

```

Figure 15: Body of *eudClassify()* routine in *sim-dir/cntrl/exp.go* of "EUDS" application

Now we see how the map *fID2CP* is used to yield the name of the EUD when the saved function ID to its *eudProcess* function is encountered. The user placed a pointer to this function in the terminal measure function's State struct, see line 66-68 of Figure 12. The **pces** code that creates a new data structure for a measurement group calls this function and puts its return string into the 'GroupDesc' attribute (see again line 169 of Figure 14).

The spreadsheet output from running this application has entries from multiple measurement groups. We can differentiate between those groups through the value in the cell under the "EUD" heading.

Experiment	MeasureName	EUD	\$crypto	\$lambda	\$pckklen	Samples	Min(msec)	Q1	Q2	Q3	Max(msec)
exp-1	end2end	EUD-1	AES-128-CBC	10000	128	1007	0.358434	0.35874	0.359079	0.359435	0.371689
exp-1	end2end	EUD-2	AES-128-CBC	10000	128	4977	0.358434	0.358809	0.35918	0.359529	0.382351
exp-1	end2end	EUD-0	AES-128-CBC	10000	128	4016	0.358434	0.358801	0.359157	0.359507	0.380874
exp-2	end2end	EUD-1	AES-128-CBC	10000	256	1007	0.377048	0.37743	0.377815	0.378197	0.405855
exp-2	end2end	EUD-2	AES-128-CBC	10000	256	4977	0.37705	0.377513	0.377944	0.379935	0.429253
exp-2	end2end	EUD-0	AES-128-CBC	10000	256	4016	0.377048	0.37748	0.377892	0.379486	0.433754
exp-3	end2end	EUD-0	AES-128-CBC	10000	512	1007	0.411875	0.41241	0.412937	0.420384	0.485423
exp-3	end2end	EUD-2	AES-128-CBC	10000	512	7018	0.411876	0.412731	0.416393	0.43506	0.701672
exp-3	end2end	EUD-1	AES-128-CBC	10000	512	1975	0.411875	0.412424	0.41296	0.421242	0.514624
exp-4	end2end	EUD-1	AES-128-CBC	10000	1024	1007	0.483936	0.484987	0.499973	0.521756	0.691711
exp-4	end2end	EUD-2	AES-128-CBC	10000	1024	4977	0.483932	0.491917	0.520545	0.58539	1.29262
exp-4	end2end	EUD-0	AES-128-CBC	10000	1024	4016	0.483932	0.491271	0.512837	0.55661	1.08776
exp-5	end2end	EUD-0	AES-256-CBC	10000	128	4016	0.359634	0.359999	0.360341	0.360687	0.383274
exp-5	end2end	EUD-1	AES-256-CBC	10000	128	1007	0.359634	0.359933	0.360265	0.360618	0.372889
exp-5	end2end	EUD-2	AES-256-CBC	10000	128	4977	0.359634	0.360005	0.360372	0.360717	0.382931
exp-6	end2end	EUD-2	AES-256-CBC	10000	256	7018	0.379448	0.379955	0.380447	0.382933	0.467208
exp-6	end2end	EUD-1	AES-256-CBC	10000	256	1975	0.37945	0.379866	0.380258	0.380891	0.414283
exp-6	end2end	EUD-0	AES-256-CBC	10000	256	1007	0.379448	0.379837	0.380224	0.380611	0.412929
exp-7	end2end	EUD-0	AES-256-CBC	10000	512	1007	0.420275	0.420828	0.421367	0.429275	0.503423
exp-7	end2end	EUD-2	AES-256-CBC	10000	512	7018	0.420276	0.421186	0.427354	0.451801	0.795453
exp-7	end2end	EUD-1	AES-256-CBC	10000	512	1975	0.420275	0.420879	0.421457	0.430616	0.531006
exp-8	end2end	EUD-2	AES-256-CBC	10000	1024	7018	0.500733	0.53505	0.655447	0.832519	2.18339
exp-8	end2end	EUD-1	AES-256-CBC	10000	1024	1975	0.500732	0.503982	0.523114	0.552344	0.855792
exp-8	end2end	EUD-0	AES-256-CBC	10000	1024	1007	0.500736	0.501864	0.51643	0.540226	0.751176

There are numeric differences between the various quartile values in a given row, but they are of such numerical insignificance that they do not appear in the table.

SJF (Shortest Job First) Application

Application structure

pces provides the 'srvRsp' class for functions that respond to service requests. When a srvRsp server has more active concurrent requests than it has resources to service them, the default behavior is to provide service to requests in first-come-first-serve order. This is common, and commonly a default, but not other queueing disciplines are sometimes used. The discipline implemented in this application is known as "Shortest Job First Pre-emptive Resume", which we will shorten to SJF. Under this discipline if a server can process n tasks concurrently, and there are $N > n$ tasks that have requested service and not yet completed it, at any time the n tasks receiving service are those n whose remaining unfilled service requirements are shortest. The collective identity of those n tasks changes of course when one of them completes its service, at which point the task that is waiting which has least remaining service requirement goes into service. When a new request arrives it may go *immediately* into service if its service requirement is smaller than that of the task in service whose remaining service requirement is largest among those in service. That task is suspended, its remaining service requirement is noted, and the new task goes into service.

One of the points of including this application is to show how a user can integrate their own implementation of the 'srvRsp' function's response to a service request.

Figure 16 illustrates the form of a CmpPtn that is replicated in this example, three times.

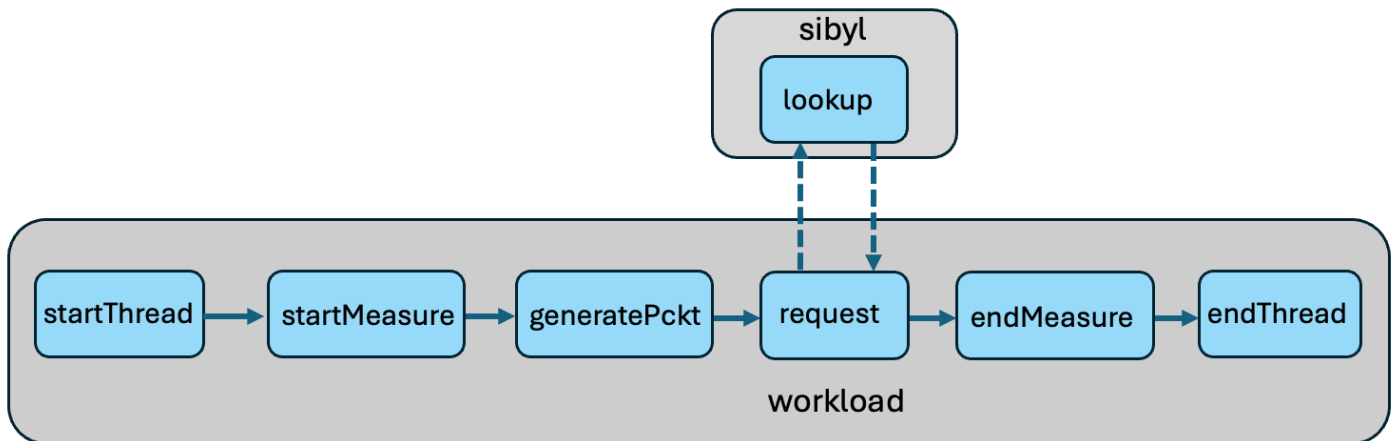


Figure 16: Example Workload CmpPtn illustrated in SJF application

The 'workload' style of CmpPtn uses functions we've seen before, with the exception that 'request' is a function of the 'srvReq' class and directs its request to the 'lookup' function (of class 'srvRsp' in the 'sibyl' CmpPtn).

Figure 17: illustrates the architecture to which these CmpPtns are mapped.

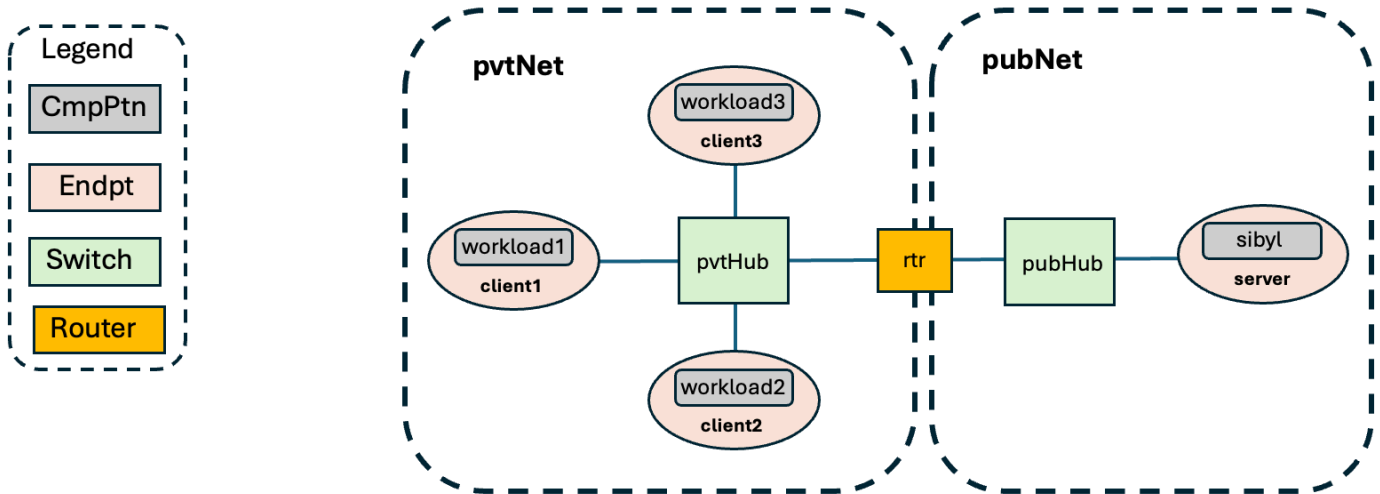


Figure 17: Architecture and CmpPtns in SJF application

This model has each 'workload' CmpPtn generate service requests via a Poisson process similar to that of the 'Queueing' and EUDs' applications. There a rate parameter λ was passed to a 'start' function, which scheduled the starting measurement of a packet generation process. The SJF application allows for different generation rates for each of the three workload CmpPtns, as illustrated in the snippets of the xlsxPCES model. The first is from the 'cp' sheet, describing the initialization of the three 'start' functions.

### start class	cmpptn	label	pcktlen	msglen	msg type	start time	data	trace	groups
start	workload1	startThread	1000	1500	measure	0	float(λ_1)		
start	workload2	startThread	1000	1500	measure	0	float(λ_2)		
start	workload3	startThread	1000	1500	measure	0	float(λ_3)		

The second is from the 'experiments' sheet, assigning values to each of these rates, for a one-run experiment.

Experiments			
name	λ_1, cp	λ_2, cp	λ_3, cp
exp-0	25.00	12.50	5.00

Application code

SJF is built with three directories, *main*, *cntrl*, and *sjf*. *main/main.go* is exactly the same as that file in the 'Embedded' application, as it requires no user supplied arguments.

ExpCntrl() illustrates another technique a **pces** model can use to identify functions of particular interest.

```

53 // ExpCntrl is called to initialize the events for the simulation
54 func ExpCntrl(evtMgr *evtm.EventManager, context any, data any) any {
55     rand.Seed(pces.GlobalSeed)
56 
```

```

57     extendSetup()
58     SaveLambda = make(map[int]float64)
59
60     // visit every CmpPtn
61     for _, cpi := range pces.CmpPtnInstByID {
62         for _, cpfi := range cpi.Funcs {
63             if cpfi.Class == "start" {
64                 // get inter-arrival rate lambda
65                 srcCfg := cpfi.Cfg.(*pces.StartCfg)
66                 param := srcCfg.Data
67                 lambda, err := strconv.ParseFloat(param, 64)
68                 if err != nil || !(lambda > 0.0) {
69                     panic("expect lambda to be positive float")
70                 }
71                 SaveLambda[cpfi.ID] = lambda
72                 arrivalTime := interArrival(SaveLambda[cpfi.ID])
73                 evtMgr.Schedule(cpfi, nil, startEnterCycle,
74                     vrtime.SecondsToTime(arrival))
75             }
76         }
77     }
78     return nil
79 }

```

Figure 18: Code body of *ExpCntrl* in *SJF* Application

Just as 'EUDs' called a routine *extendSetup* to establish some linkage between user-defined functions and **pces** internals, so does the call at line 57. We will return to that. In line 58 we create a map that will be indexed by a function identity (that of a 'start' function) and return a request arrival rate associated with that function.

The loops rooted in lines 61 and 62 visit every function of every CmpPtn in the model, and in line 63 identify functions in the 'start' class. We know each such function has been configured with a request rate in the 'Data' field of its configuration, extract it, save it to the *SaveLambda* map in line 71, and schedule an immediate call to *startEnterCycle* in line 72. This version is almost identical to that in *euds/sim-dir/cntrl/exp.go*, save that the 'SJF' version skips lines 99-115 of Figure 13, as these establish the per-EUD sampling distribution. Another difference is that the 'SJF' version does not test against the number of samples made. In order to keep all three workload CmpPtn generators always operating concurrently, the simulation run is terminated after a declared simulation time. Still, as with the 'EUDs' version, the net effect is to launch a self-scheduling process that generates requests stochastically as a Poisson process.

The code of *extendSetup* bears examination.

```

22 // user extension of core methods for srvRsp and finish function classes
23 func extendSetup() {
24     // visit every CmpPtn
25     for _, cpi := range pces.CmpPtnInstByID {
26         // find list of functions in group "SJF", if any
27         funcLists, present := cpi.FuncsByGroup["SJF"]

```

```

28     if present {
29         // visit every function in that group, in this CmpPtn
30         for _, cpfi := range funcLists {
31             // we need a function of the srvRsp class
32             if cpfi.Class != "srvRsp" {
33                 continue
34             }
35             // found one
36             cpfi.AddStartMethod("sjf", sjf.SrvRspSJF)
37         }
38     }
39     funcLists, present = cpi.FuncsByGroup["EndMsr"]
40     if present {
41         // there is only one function in this group
42         for _, cpfi := range funcLists {
43             if cpfi.Class == "measure" {
44                 msrState := cpfi.State.(*pces.MeasureState)
45                 msrState.Classify = CPClassify
46             }
47         }
48     }
49 }
50 }
51
52 // CPClassify returns the CmpPtn name of execution thread source
53 func CPClassify(visited []int) string {
54     return pcres.CmpPtnFuncInstByID[ visited[0] ].PtnName
55 }

```

Figure 19: Code body of extendSetup in SJF Application

extendSetup() needs to find the 'srvRsp' response function. Now we do know the CmpPtn name and function label and so could initialize the variable *cpfi* pointing to that function's configuration just as we did in lines 31-32, in Figure 12. looking up the 'startThread' function of the 'HMI' CmpPtn, but have chosen to include the loop rooted in line 25 to illustrate how we can use the 'Groups' list that every function is configured to have. In this model we configured the server function to be part of a group we defined, "SJF". See the snippet of *xlsxPCES* spreadsheet below, from the 'cp' sheet.

### srvRsp class	cmpptn	label	(timingcod	(timingcod	directprefix	trace	(msg2mc)	(msg2m	groups
srvRsp	SJFServer	lookup	request1	sjf1			request1	sjf	SJF
			request2	sjf2			request2	sjf	
			request3	sjf3			request3	sjf	

During startup **pcres** constructs for each CmpPtn a map, indexed by string, associated with a list of pointers to that CmpPtn's functions who have the index key listed in their 'Groups' list. For this example there is one. The advantage of this technique as compared to that shown in Figure 12 is that we do not need to hard-wire the names of the CmpPtn and function to identify the data structures of interest. Also note lines 42-47 find functions from class 'measure', and configure them to call routine *CPClassify()* to provide a name to a group of

measurements that share a path. Line 54 uses our knowledge that the first step in the path will point to a function whose data structure can tell us (through field 'PtnName') the name of the CmpPtn the function sits in.

Line 36 is absolutely critical to what we're doing. Recall from companion document [PCES-Overview](#) that the response a function makes to receipt of a message is obtained by translating the message's message type into a 'method code', and then the method code is used as an index into a dictionary that is specific to that function, which returns a pointer to the subroutine to call to handle that message. Line 36 identifies the method code---'sjf'---and the subroutine---sjf.SrvRspSJF to call when sjp is the method code. The srvRsp configuration above says to map message types 'request1', 'request2', and 'request3' to that method code (and to use timing codes 'sfj1', 'sfj2', and 'sfj3' to look up service requirements as part of computing the response). The message type on a message from a function of class 'srvReq' to a function of class 'srvRsp' is taken from the former function's initialization, of the 'SrvOp' field, listed below for each of the three 'srvReq' functions.

###	srvReq class	cmpptn	label	bypass	trace	(msg' (msg2	srvcpl	srvlabel	srvop	rspop	(msg2' (msg	groups
	srvReq	workload1	request				SJFServer	lookup	request1			
	srvReq	workload2	request				SJFServer	lookup	request2			
	srvReq	workload3	request				SJFServer	lookup	request3			

Since the `sjf.SrvRspSJF()` function is called instead of the default response function (`pces.srvRspEnter()`), transparently to the 'srvReq' function that called it, regardless of how it manages allocation of server resources it has to generate a response that is syntactically and semantically identical to that of `pces.srvRspEnter()`. In addition, `sjf.SrvRspSJF()` needs to include all of the shortest-jobs-first logic. That all is too detailed to be of interest in this particular document (but the interested reader can view github.com/iti/pcesapps/sjf:sim-dir/sjf/sjf.go), but suffice to say a sensible approach for a user is to copy the function being replaced, and edit it to include the desired functionality, while retaining the code that reports service completions.

For the given experiment described by the 'experiments' sheet illustrated in the introduction to the 'SJF' application (and a simulation run time of 1000 virtual seconds) the results given are illustrated below.

Experiment	MeasureName	\$lambda1	\$lambda2	\$lambda3	Samples	Mean(msec)	StdDev
exp-0	response1	25	12.5	5	24848	16.958	4.53677
exp-0	response2	25	12.5	5	12645	40.3479	20.1037
exp-0	response3	25	12.5	5	5150	168.209	116.032

Not shown in any of the other data we've supplied for this application is that the service requirements for workload1 is 10 msec, for workload2 is 20 msec, and for workload3 is 50 msec. This helps us decipher the results above. For each of the workloads the bottleneck step is the service. The request generation rates for each workload is 25% of that workload's service rate, which means we should achieve stationary queue behavior. The mean response time of the workload with shortest service times is not terribly far off its mean service time, 16.9 msec compared to 10 msec. However, we see that the mean response times for the other two workloads are significantly larger than the mean service times, a result of them having to queue waiting for shorter requests from other workloads to complete. The difference is even more marked when we double the request rate for workload1 while leaving the other workloads the same.

Flows Application

Application structure

As described in the **mrnes** documentation, a unique feature of **mrnes** is its support for concurrent simulation of network components at different levels of resolution. The point of the design is to enable a modeler to include the impact that computation and communication *other* than that of particular interest has on the workflows of interest, by including their use of network and computational resources, against which the foreground activity competes.

Flows can be introduced to consume bandwidth of interfaces and networks, and so slow down communication. A flow has a source endpoint and destination endpoint, and the configuration specifies the bit rate that traffic flows consume from source to destination, and whether the simulator represents that flow as a sequence of explicitly generated and explicitly transmitted frames, or whether the bit rate is *interpreted* in such a way as to model the impacts that explicitly simulated frames at that bitrate would have on the waiting time that foreground traffic frames undergo as they transit the network.

Figure 20 illustrates the flows application topology, and computational pattern.

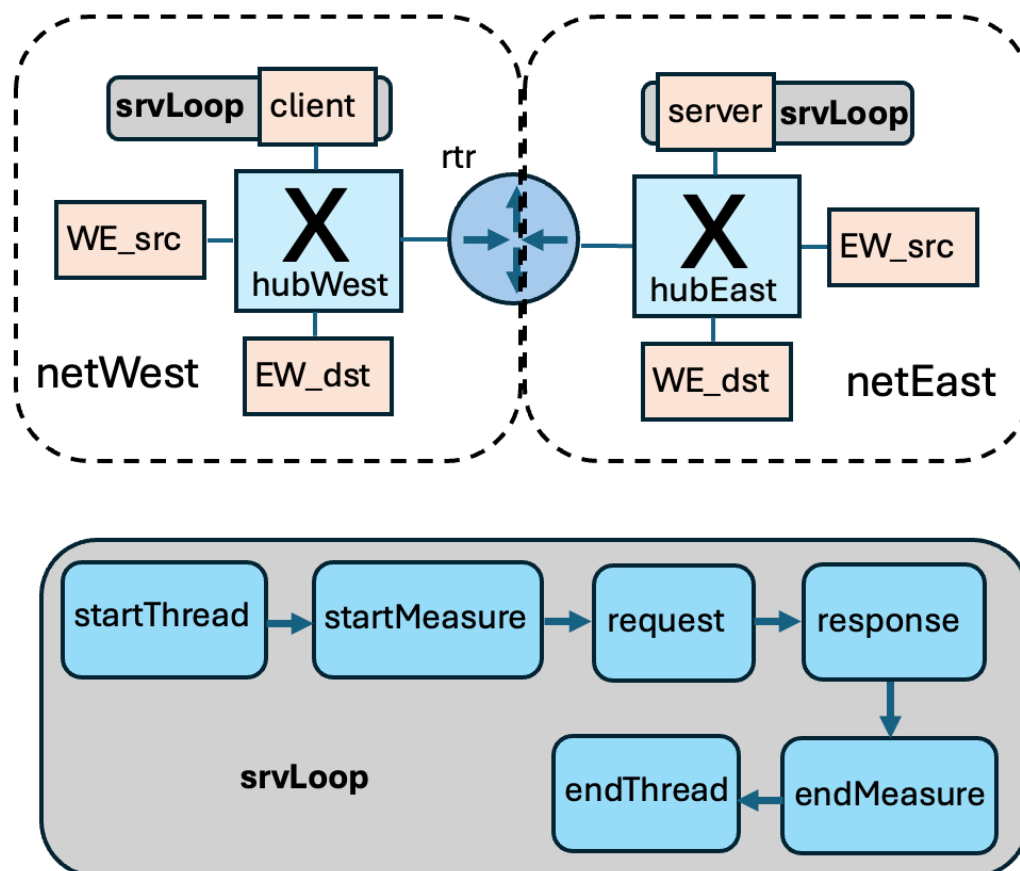


Figure 20: Flows application

The topology has networks 'netWest' and 'netEast' bridged by router 'rtr'. Each network has an endpoint designated as being a source or destination for one of two flows: 'flowEW' originating at host 'EW_src' and directed to 'EW_dst', or 'flowWE' originating at host 'WE_src' and directed to host 'WE_dst'. Each network has a hub switch. Network 'netWest' has a host 'client' home to a `pces**` function that generates requests directed to a `pces` function on host 'server' in 'netEast'. Figure 20 also illustrates the computational pattern 'srvLoop' that organizes all the functions involved in this request-response loop. The only function shown that is *not* mapped to host 'client' is 'response', which is mapped to host 'server'.

The diagram below illustrate how the flows are defined in `xlsxPCES` (in the 'topo' sheet).

Flows						
### name	src endpt	dst endpt	mode (pckt, elastic-flow, inelastic-flow)	reqRate	framesize	groups
flowWE	WE_src	WE_dst	str(\$mode)	float(\$Lambda)	1000	
flowEW	EW_src	EW_dst	str(\$mode)	float(\$Lambda)	1000	

Each flow has a unique name, and for each the name of the endpoints serving as source and destination are given. The mode of the flow is either packet-oriented or one of two models of flow-oriented (explained in [mrnes.pdf](#)) is an experimental variable that in the experiments we report takes values 'pckt' and 'inelastic-flow'. 'reqRate' is another experiment variable, that we vary to change the requested bit rate of the flow (in Mbits/sec).

The statement of initiation variables for the 'startThread' function is shown below.

### start class	cmpptn	label	pcktlen	msglen	msg type	start time	data	trace	groups
start	reqLoop	startThread	900	1000	measure	0	float(\$msrLambda)		

The key points are that the framesize of the message is 1000 bytes, the loop that starts requests being generated begins at time 0, and that the 'data' is an experiment variable. When quantified it is interpreted as the rate of service requests per second.

Examination of the topology in Figure 20 tells us that traffic from the foreground process generated by 'reqLoop' competes for service with the flow originating at 'WE_src' at the egress interface from hubWest to rtr, and that traffic from the response traffic carried by 'reqLoop' competes for service with the flow originating at 'EW_src' at the egress interface from hubEast to rtr. As a result we expect to see that as we increase the requested flow rate we will see an increase in the round-trip latency of requests for service created and measured within the 'srvLoop' traffic.

The set of experiments described in the 'experiments' sheet vary the rate of the two flows, comparing the latency of a foreground probe initiated by the client when the background flows are packet oriented, and when they are expressed as bit-rates

Experiments			
#name	\$Lambda,topo	\$mode,topo	\$msrLambda, cp
exp-0	10	inelastic-flow	625
exp-1	10	pckt	625
exp-2	20	inelastic-flow	625
exp-3	20	pckt	625
exp-4	30	inelastic-flow	625
exp-5	30	pckt	625
exp-6	40	inelastic-flow	625
exp-7	40	pckt	625
exp-8	50	inelastic-flow	625
exp-9	50	pckt	625
exp-10	60	inelastic-flow	625
exp-11	60	pckt	625
exp-12	70	inelastic-flow	625
exp-13	70	pckt	625
exp-14	80	inelastic-flow	625
exp-15	80	pckt	625

The bottleneck in the example is the (common) interface bandwidth, set to 100 Mbs. Working through the math, a rate of 625 foreground packets per second translates into 5 Mbit/s demand, so varying \$Lambda between 10 Mbit/s and 80 Mbit/s varies the demand for bandwidth on the interfaces between the switches and rtr from 15% to 85%.

The egress interface from hubWest to rtr will be presented with asynchronous traffic from client (the foreground traffic) and WE_src, the source of the West-to-East background flow. We expect to see that the background traffic induces queueing at that interface for the foreground traffic. Similarly, in netEast, the East-to-West flow will induce queueing for the foreground traffic returning from server, through the egress interface from hubEast to rtr. We can see these effects in the results of experiments run in the csv file of experimental results. To compare the predictions of the model using flows created from packets and that of the model created from flows specified only by bit-rate we compare the Mean column of the row with \$mode==inelastic-flow with that of the row with \$mode==pckt for the same value of \$Lambda. We can see from the table that there is close but not perfect correspondence, but that the bit-rate model closely tracks the growth in Mean of the packet model as \$Lambda grows.

Experiment	MeasureName	\$Lambda	\$mode	\$msrLambda	Samples	Mean(msec)	CI	#Events
exp-0	loop	10	inelastic-flow	625	614	1.4137	0.00304642	584144
exp-1	loop	10	pckt	625	614	1.412	0.0029947	1431744
exp-2	loop	20	inelastic-flow	625	614	1.42229	0.00383627	585529
exp-3	loop	20	pckt	625	614	1.42256	0.00381351	2283982
exp-4	loop	30	inelastic-flow	625	614	1.42959	0.0041674	586918
exp-5	loop	30	pckt	625	614	1.4348	0.00462077	3128766
exp-6	loop	40	inelastic-flow	625	614	1.43832	0.0044422	588123
exp-7	loop	40	pckt	625	614	1.44366	0.00504054	3974773
exp-8	loop	50	inelastic-flow	625	614	1.45356	0.00489572	589491
exp-9	loop	50	pckt	625	614	1.45343	0.00517884	4826897
exp-10	loop	60	inelastic-flow	625	614	1.46021	0.00527398	590733
exp-11	loop	60	pckt	625	614	1.46412	0.0056142	5672903
exp-12	loop	70	inelastic-flow	625	614	1.47155	0.00546598	592180
exp-13	loop	70	pckt	625	614	1.48405	0.00657422	6527661
exp-14	loop	80	inelastic-flow	625	614	1.48744	0.00586995	593767
exp-15	loop	80	pckt	625	614	1.50738	0.00747023	7393390

Illustrate the performance gain of the bit-rate model we also include in the last column the number of discrete events that were executed by the simulator to establish the reported results. Here we see that the number of events executed by the bit-rate model remains relatively constant as λ grows, but (of course) the event count for the packet model grows in proportion to λ , and in all cases is significantly larger than the event count for the bit-rate model, ranging from x2.5 at $\lambda=10$ to x12.5 at $\lambda=80$.

AS Topo Application

The AS topo application is included to highlight two features of **pces/mrnes** we've not yet highlighted. Figure 21 below illustrates the network topology, and the computational pattern that is mapped to it.

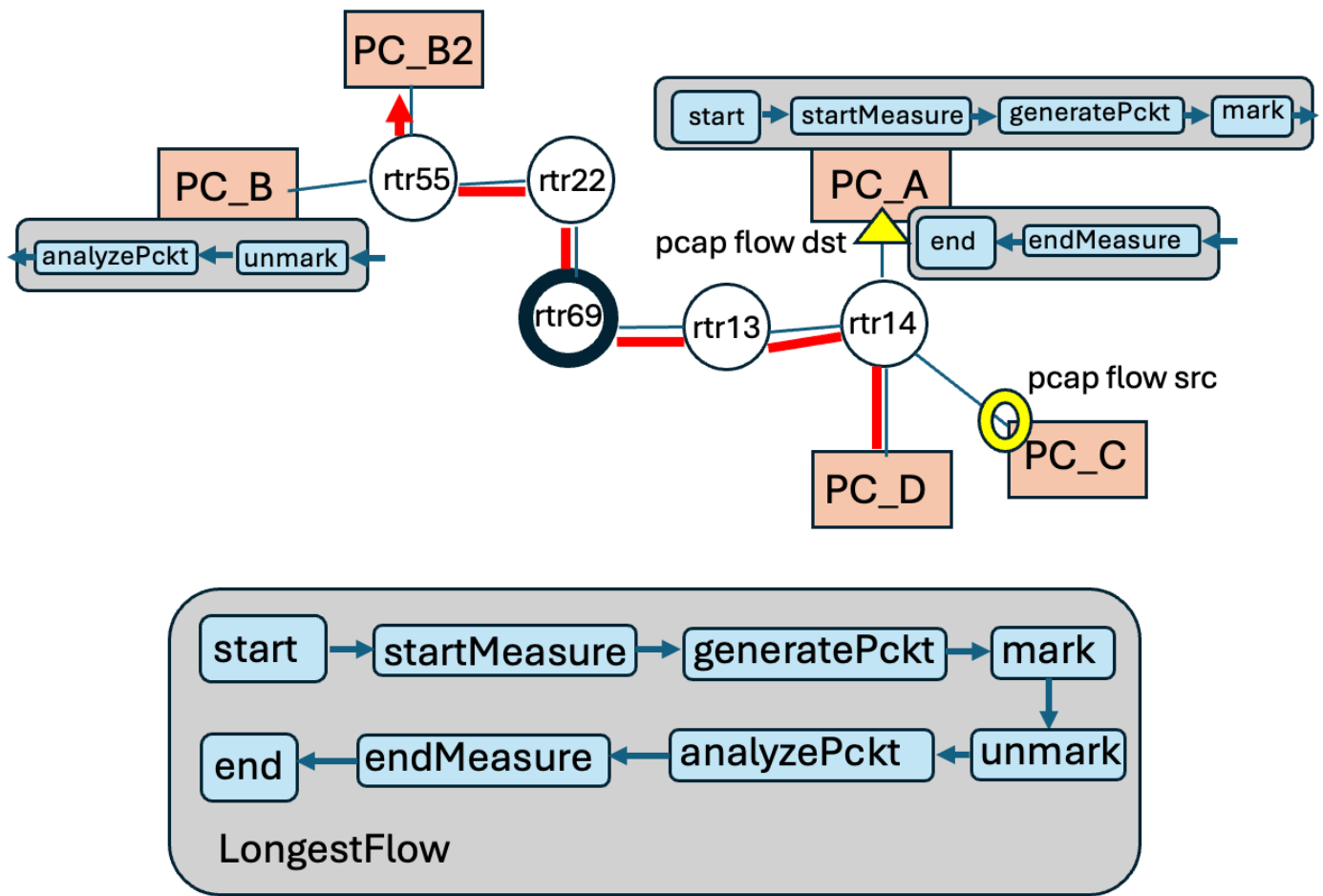


Figure 21: AS Topo application

In this model PCAP packets are introduced into the model. The configuration we outline below specifies that a file of PCAP packets be read, and introduced at interfaces that have IP addresses corresponding to the source IP addresses of the packets. In this particular model all of the packets have the same source IP address and they all have the same destination IP address. The configuration specifies that the source IP is bound to PC_C's interface and the destination IP is bound to PC_A's interface. Packets from this feed are introduced into the model at PC_C, at simulation times derived from the times on the PCAP packets themselves, and the transition across the network to PC_A is simulated like any other packet in the simulation would be. Upon receipt of a PCAP packet at the PC_A interface the computational function 'Start' triggers an execution of the CmpPtn 'LongestFlow'.

The topology includes five routers. Router rtr69 is highlighted to point out that it includes additional hardware that selectively performs some computation on a packet. The CmpPtn 'LongestFlow' looks like others we have already seen, but with the addition of two new functions, one labeled 'mark', the other labeled 'unmark.' Both functions are of the 'metadata' class. 'mark' includes a metadata variable named 'compute' (with value 'hash') that is attached to the message traversing the CmpPtn message and is included with the **mrnes** message traveling from the source PC_A to destination PC_B across the network. As that message passes through rtr69, the router recognizes the meta data variable as one it is programmed to react to, and delays passage of the packet to account for the additional processing being modeled. Leaving rtr69 the message continues to carry the same metadata, but on reaching PC_B and executing function 'unmark' that metadata variable is removed.

This means that when the message passes through 'analyzePckt' and returns to PC_A, on passing through rtr69 again no additional processing is modeled.

This model also includes specification of a background flow originating in PC_D and directed through the network to PC_B2. This flow may impact packets sent from PC_A to PC_B, but only at the egress interface of rtr14. Thereafter packets that the flow represents would be interleaved through queueing with packets from PC_A at rtr14.

Let us now examine pieces of the xlsx description that relate to the newly introduced functionality. The 'topo' sheet is standard, it describes the devices and their interconnections, and the flow just as we have done in other examples.

We do not require that IP addresses contained within the simulation model match exactly to IP addresses found on the PCAP packets that are introduced. So instead, assuming that some external analysis of the PCAP source identifies all the IP headers that might possibly be carried into the simulator, we use the 'ipmap' sheet to describe the mapping from the external IP addresses to what they correspond to in the simulation model. The 'ipmap' sheet from this example is given below.

Device						
### name	network	internal IP	external IP			
PC_C	net_A	10.0.1.1	192.168.66.33			
PC_A	net_A	10.0.1.2	192.168.66.34			
Network						
### name	internal CIDR	external CIDR				
Feed						
### name	active	src type (file, unix-socket, net-socket)	src spec (file name or port number)	time (clock, packet)	dilation factor	first arrival (sec)
Feed1	TRUE	file	pckts.pcap	packet	2.5	0.75

The Device block associates expected external IP addresses with the interface and its IP address that represent it. We identify the interface by specifying the device that hosts it, and the network it faces. A result of processing this information as the model is built is that that interface is found in the model (it is an error if that interface doesn't exist) and the given internal IP address is assigned to it.

The Network block associates external and internal CIDR blocks associated with a network in the model. Such mappings are not needed in this example.

The Feed block describes how PCAP packets can be introduced into the model. This example has only one feed, in general there may be multiple feeds. The 'active' column enables one to toggle a feed on or off between experiments. The 'src type' indicates whether the PCAP packets are delivered by reading a file, reading them from a socket declared as a Unix-file, or through a network socket. The 'src spec' column gives the PCAP file name for the first source type, the (absolute) Unix file name for the second source type (a file that is accessible both to the process that writes packets to file and the simulator that reads them), and gives the port number on the simulator's host in the case of the third source type.

The 'time', 'dilation factor' and 'first arrival' columns describe how to translate times contained in an arriving PCAP packet into virtual simulation times. It is, (well, actually, will be shortly) possible to run the simulator in real-time, meaning that the simulation clock advance is directly tied to wallclock time advance, and the virtual time given to an arriving PCAP packet is the virtual time of the simulator at the wallclock instant of arrival. One selects the code 'clock' in the 'time' column for this option. Otherwise the 'packet' option is expected, which

means that time-stamps associated with the PCAP packet are transformed into virtual times.

PCAP packet times are (after a tiny bit of preprocessing before delivery to the simulator) in units of microseconds in the Unix epoch, time zero being midnight of Jan 1, 1970. The value in the 'first arrival' column specifies the virtual time to be given to the first packet that arrives in the feed. The time stamps of a subsequent arrival is obtained by computing how long after the first packet it arrives, scaling that by multiplying this difference by the dilation factor given in the 'dilation' column, and adding that to the virtual arrival time of the first packet. Thus the virtual time of a packet is a linear transformation of the time of its measurement.

The 'cp' sheet references two new function classes 'feed', and 'metadata' declared in the Patterns block of that sheet.

Patterns					
### name	type	func class	func label	SrvOp	Service CP, Service Func
LongestFlow		feed	start		
LongestFlow		measure	startMeasure		
LongestFlow		processPckt	generatePckt		
LongestFlow		metadata	mark		
LongestFlow		metadata	unmark		
LongestFlow		processPckt	analyzePckt		
LongestFlow		measure	endMeasure		
LongestFlow		finish	end		

The 'start' function has class 'feed', designed to start execution on receipt of a packet introduced by a PCAP source. The 'mark' and 'unmark' functions have class 'metadata', and serve to modify a meta data dictionary that can be applied to a CmpPtn message, and be visible at the **mrnes** layer as a network packet supporting communication between CmpPtn functions is carried.

Details about what these functions are configured to do are specified in the function's initialization blocks.

### feed class	cmpptn	label	Src Feed IP	Dst Feed IP	msg type	data	trace	groups
feed	LongestFlow	start		10.0.1.2:*	default			
### metadata	cmpptn	label	remove	(add) meta data name	(add) meta data value	trace	groups	
metadata	LongestFlow	mark		compute	hash			
metadata	LongestFlow	unmark	compute					

We initialize the 'start' function with specification of the PCAP feed that it accepts. Here we specify requirements on the IP header of PCAP packets to be delivered to this function. Both an IP address and port number are given in the string of the form "IP:port". The IP number is one that should appear in the 'internal IP' column of the 'ipmap' sheet. For both IP and port number we allow a wildcard specification "*", with one exception, the IP address given for the destination IP cannot be a wildcard. For a PCAP sourced packet to be delivered to the function, the IP header of that packet must match the specifications in this table whenever the specification is not a wildcard. For the configuration shown that means that every PCAP packet whose destination IP maps to the simulation's IP address of 10.0.1.2. Together with the information from the 'ipmap', this means that every PCAP packet that is introduced whose native destination IP address is 192.68.66.34 will be accepted at device PC_A and delivered to LongestFlow's 'start' function.

Another feature of **mrnes/pces** modeling is illustrated in this example. We were motivated by a use case where certain packets receive specialized processing in routers, which adds latency, and may also lower throughput. We need to have a way of specifying which packets are to receive this processing, and which devices provide it. We do this by associating with each CmpPtn and each network message a dictionary that maps strings to strings. We call this meta data. We need a way of introducing meta data to selected packets, and a way of recognizing the packets and including the added processing cost. The 'metadata' function class is included to enable a modeler to add or delete meta data on CmpPtn messages that pass through it. The initialization for this function includes a list of strings (in the 'remove' column), and a mapping from meta data key values (in the 'meta data name' column) to values (in the corresponding 'meta data value' column). Entering a function of the metadata class, the message's meta data dictionary is stripped of entries whose name value appears in the 'remove' list, and then adds the names and corresponding values specified in the add dictionary. When a CmpPtn message is carried through the network that dictionary is visible to network devices.

The meta data name 'compute' is one that **mrnes/pces** recognizes without further user specification.

We specify the devices that provide additional computation and the cost of that computation in the 'execTime' sheet.

CPU Entries				
### model	operation	pcktlen	exec time	
Dell	packet-generation	50	10	
Dell	packet-generation	1500	10	
Dell	packet-analysis	50	10	
Dell	packet-analysis	1500	10	
Accelerator Entries				
### model	operation	pcktlen	exec time	
Router Entries				
### model	operation	pcktlen	exec time	bandwidth
Cisco ASR	route	50	40	1000
Cisco ASR	route	1500	40	1000
Cisco ASR extended	route	50	40	1000
Cisco ASR extended	route	1500	40	1000
Switch Entries				
### model	operation	pcktlen	exec time	bandwidth
Function				
### model	operation	pcktlen	exec time	
Cisco ASR extended	compute	50	1000000	
Cisco ASR extended	compute	1500	1000000	

The addition to examples we've seen before is the 'Function' block. These specify the meta data names on packets passing through a device that trigger added delay costs. Just as operation and function costs depend on the device model and packet length, so too do the additional function costs. In this example only rtr69 is intended to apply those functions, and so we give it a model type that is unique, here, 'Cisco ASR extended'. In

any case it makes sense to assume that devices with this additional capability are different and have a different model specification.